



OECD Science, Technology and Industry Working Papers
2025/21

Innovation policy
transformed? Unveiling a
new paradigm through
natural language processing

Jan Einhoff,
Caroline Paunov

<https://dx.doi.org/10.1787/5ee60cb5-en>

OECD Science, Technology and Industry Working Papers

OECD Working Papers should not be reported as representing the official views of the OECD or of its member countries. The opinions expressed and arguments employed are those of the authors.

Working Papers describe preliminary results or research in progress by the authors and are published to stimulate discussion on a broad range of issues on which the OECD works. Comments on Working Papers are welcomed, and may be sent to Directorate for Science, Technology and Innovation, OECD, 2 rue André-Pascal, 75775 Paris Cedex 16, France.

Note to Delegations:

This document is also available on O.N.E under the reference code:

DSTI/STP/TIP(2023)10/REV1/FINAL

© OECD 2025



Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence

(<https://creativecommons.org/licenses/by/4.0>).

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: In the event of any discrepancy between the original work and the translation, only the text of original work should be considered valid.

Adaptations – you must cite the original work and add the following text: This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement. You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work.

Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Innovation Policy Transformed? Unveiling a New Paradigm through Natural Language Processing

Jan Einhoff and Caroline Paunov

Innovation policy is changing. No longer focused solely on growth and competitiveness, today's strategies increasingly address societal challenges such as the green transition, digitalisation, and social inclusion. This paper provides the quantitative evidence of this shift, analysing STI strategies from 24 OECD countries using natural language processing techniques. It finds that goals linked to societal impact now make up over 40% of strategies' topics — more than twice that of traditional objectives. New policy approaches, including joint governance, foresight, and agility, are also gaining ground. This change in innovation policy is mainly driven by the prominence of green transition goals in strategies (18.6%), followed by social and regional inclusion (9.9%) and digitalisation (9.5%). New STI policy approaches are also prominently featured with references to joint governance (5.4%), policy agility (5.2%) – such as foresight and evaluation – and directional policy (2.3%).

Key words: innovation and technology policy, natural language processing, topic modelling,

JEL codes: O38, O31, O25, Q55, L52, C88

Acknowledgments

The authors would like to thank Sandra Planes Satorra, Luke Mackle, Paula Puchol Ledo, Nikolas Schmidt, and Isabella López Trejos, from the OECD-TIP team, for providing valuable inputs to the paper.

The authors would also like to express their sincere gratitude to the TIP delegates for their insightful comments during the development of this report, as well as for their support in identifying and validating the documents used for the analysis.

The authors also extend their gratitude to the speakers and participants of workshops organised by the OECD Working Party on Innovation and Technology Policy (TIP) on the topics discussed in this paper in 2022 and 2023.

Additionally, the authors are grateful to Andrew Paterson (OECD), Alexander Lembcke (OECD) and Rudiger Ahrend (OECD) for their in-depth review of the paper.

Table of contents

Acknowledgments	4
Executive summary	6
1 Introduction	8
2 Conceptual framework	11
3 Data and empirical strategy	13
4 Findings	19
5 Conclusions	27
References	28
Annex A. Pre-processing and cleaning steps for the topic modelling	31

Tables

Table 1. Descriptive statistics for the final database of STI policy strategy documents	14
Table 2. Shares of coverage of all topics, themes and categories in the corpus	17
Table 3. Share of Innovation policy 3.0 across topic number and sample variation	22
Table 4. Share of green transition goals across topic number and sample variation	23
Table 5. Distribution of shares of innovation policy 3.0 topics across countries	24
Table 6. New innovation policy approaches	24
Table 7. Examples of new innovation policy approaches from the corpus	25

Figures

Figure 1. Exclusivity and semantic coherence for distinct numbers of topics	16
Figure 2. Shares of topic categories across strategies in 24 OECD member countries	19
Figure 3. Share of topics categories across model variations	20
Figure 4. Cross-country distribution of the shares of traditional innovation policy goals	21
Figure 5. Cross-country distribution of the shares of innovation policy 3.0 goals	22

Executive summary

A paradigm shift for STI policy?

In recent years, STI policies have come under increasing pressure to address a growing set of complex challenges, including the digital transformation, the push for environmental sustainability, and the response to the COVID-19 pandemic. These challenges carry significant social dimensions—such as the risk of labour displacement from advanced digital technologies and the rising frequency of extreme weather events—that are promoting a shift in the rationale for STI policy intervention beyond exclusively competitiveness and productivity.

This evolving approach, where traditional objectives are intertwined with new priorities, is referred to in the STI policy literature as ‘Innovation Policy 3.0’. According to this framework, societal goals—such as inclusiveness and environmental sustainability—are placed at the centre of STI efforts. Innovation is no longer presumed to be inherently beneficial; instead, it is assessed based on its broader social and environmental impacts.

However, there is still limited evidence on the extent to which countries have adopted this new strategic outlook and implemented it in practice—insights that are essential for assessing the current state of STI policy and guiding its future direction.

Leveraging topic modelling for exploring STI policy strategies

Capturing the global shift in STI policy is difficult with traditional qualitative methods due to the volume of policy documents. Topic modelling offers a scalable solution by identifying patterns and themes across large text collections for expert analysis.

Using this approach, a study of 313 STI strategies—policy frameworks outlining national priorities, goals, and actions in STI from 24 OECD countries, validated by national experts—examined whether the stated policy goals reflect a shift toward Innovation Policy 3.0 and support the adoption of new implementation approaches.

The approach identified 35 distinct “topics,” where each topic represents a group of words that frequently appear together and capture a coherent theme within the documents. This allowed for analysing how often and where these themes occur across the entire collection of policy texts.

Green and inclusion goals propel Innovation Policy 3.0 over digital

STI strategies from 24 OECD countries reveal a strong move toward Innovation Policy 3.0 goals, which represent a 41.7% share of the 35 topics that were identified. While traditional objectives such as research, business innovation, and productivity remain important (18.6%), they are now clearly secondary, or increasingly pursued in conjunction with new imperatives for STI policy.

The green transition is the main driver of this shift (18.6% of topics). Green transition-related content significantly focuses on energy generation, CO₂ emissions, and efficiency, accounting for 7.9% of the total mention of topics; sustainable industry accounts for 3.2%; and clean mobility accounts for 3%. However, biodiversity, circular economy, and climate resilience are less prominent, suggesting that these issues have yet to gain traction in national STI policy documents relative to other green transition topics.

Social and regional inclusion also emerge as a priority (9.9% share of the identified 35 topics), ahead of other Innovation Policy 3.0 dimensions like health, knowledge, and technology (3.8% combined). Most inclusion content focuses on employment and territorial disparities, while broader societal aspects (e.g., citizen engagement, ethics, and public trust in technology) remain underrepresented.

Digital and technological transformation goals make up 9.5% of strategy content (roughly half the focus given to green priorities). The relatively lower share may reflect the maturity of digital policy, where issues such as competition, ethics, and infrastructure are increasingly addressed outside STI strategies. Meanwhile, green STI requires stronger policy intervention and is thus more prominent in current strategic thinking.

New policy approaches are gaining ground

Strategies also reflect growing attention to how STI policy is implemented. New approaches to STI policy – which are included as ‘topics’ for the purpose of the analysis – account for 12.8% of the total topics in the corpus. This trend shows STI strategies not just redefining what STI is for, but also how it should be governed and delivered.

These key emerging approaches include joint governance mechanisms (5.4%), including vertical coordination and stakeholder consultation; policy agility (5.1%), with focus on monitoring, foresight, and flexible adaptation; and directional policy tools (2.3%), to steer innovation toward societal outcomes.

1 Introduction

In recent years, there has been increasing societal pressure on science, technology, and innovation (STI) to accelerate the digital and green transitions. The swift progress of digital technologies, such as AI-enabled tools, automation, and social media, has ignited debates over their potential adverse socio-economic effects. There is growing policy and academic focus on how governments can steer digital technological developments in directions that are beneficial to society (Acemoglu and Johnson, 2023^[1]), which follows from a growing recognition following the COVID-19 pandemic of how STI can support social and economic resilience beyond the traditional domains of competitiveness (Paunov and Planes-Satorra, 2021^[2]).

Addressing the complex transitional challenges that societies face implies a major shift of innovation policy moving from the first generation, to the second and third. This evolution reflects changes in external drivers, scope, directionality and objectives, progressing from a focus on basic research to the invention of “science policy” and ultimately to the emergence of new policy imperatives (Grillitsch et al., 2019^[3]). Systematic empirical evidence to date on where OECD countries’ STI policy goals stand vis-à-vis the new STI policy paradigm is limited.

Despite widespread enthusiasm about STI’s role in addressing complex societal transitions, it remains unclear whether the content of national policy frameworks that set national priorities, goals, and actions in STI has substantively changed in response. This paper uses topic modelling to analyse 313 STI strategies across 24 OECD countries. Validated by national experts and active in 2021, these strategies are examined to assess whether the goals they articulate reflect a shift toward new objectives and in the role of STI and whether new approaches to policy design and implementation emerge.

The contribution of this paper to the literature is twofold. It is the first to provide systematic evidence on whether and how STI policy strategies reflect a new societal paradigm. This validates a large body of conceptual work on new STI policies (Diercks, Larsen and Steward, 2019^[4]; Asheim, Moodysson and Tödtling, 2011^[5]; Mazzucato, 2016^[6]). Second, it demonstrates the usefulness of topic modelling for the toolbox of STI policy analysis and how to address core methodological challenge these as other tools also have. In this way, the paper anticipated the work being done in the context of the European Commission-OECD STIP Compass initiative, which has since begun integrating LLM-driven tools for researchers to use on the large corpus of policy documents the STIP Compass contains.

Critical to the approach is methodological validation to control for sample selection. The challenge intensifies for international innovation policy documents due to variations in how countries address innovation. The number and length of STI strategy documents vary significantly across countries. Not all countries have standalone innovation strategies; some incorporate innovation within broader economic and industrial strategies, necessitating the inclusion of different documents in the analysis. Sample biases are addressed by (i) obtaining validation by country policy experts to ensure that different country practices are reflected in the sample, and (ii) conducting various sensitivity tests on the sample to assess the robustness of the overall evidence.

Subjectivity biases may also affect the analysis through the selection of labels for identified themes and the determination of the number of themes. These biases are addressed by employing ChatGPT for systematic labelling. As to the choice of topics, exclusivity and semantics tests are applied to identify the optimal and cohesive number of topics, minimising the risks of excluding relevant themes. In addition, the number of topics is also varied to test the robustness of results and identify on that basis dominant themes.

Seven core findings of the analysis are the following: First, evidence from a sample of 24 OECD countries confirms that the majority has embraced the new policy paradigm, with Innovation Policy 3.0 goals representing 41.7% of the topics identified in the corpus of documents. There remains a solid focus on traditional core issues of research and business matters connected to building productivity and growth (18.6%), but at less than half the importance of Innovation Policy 3.0 goals. Moreover, given strategies' traditional focus on formulating policy objectives rather than their implementation, that 12.8% of the topics in the corpus deal with new approaches to innovation policy is remarkably high. It testifies to growing awareness on the implementation challenges with reaching Innovation Policy 3.0 goals.

Second, the main driver of Innovation Policy 3.0's prominence in strategies is the green transition, which represents 18.6% of the 35 topics. Energy generation sources, CO₂ emissions, and efficiency are a predominant theme, accounting for 7.9% of the topics. Sustainable industries (3.2%) and clean mobility (3.0%) are also targets. Absent from key themes are biodiversity, circular economy and climate resilience. The themes reflect lower prioritisation of certain themes in overall green transition-related policy debates. There are also differences across countries in the emphasis set. For instance, Japan's strategic documents emphasise alternative power generation (7.6% vs. 2.3% for the average), a response to the Fukushima nuclear accident of 2011.

Third, digital and technological transformation goals represent about half the share of green transition goals (9.5%). The strong emphasis on the green transition compared to digitalisation likely results from the needed push in green STI. Due to swift progress on digitalisation, by contrast, policy actions beyond STI on such matters as competition, ethics and infrastructures have been in high demand.

Fourth, social and regional inclusion is about as prominent as digital and technological transformation goals (9.9%), ahead of health, knowledge, and technology (3.8%), the last and final theme within Innovation Policy 3.0. Within the inclusion theme, questions relate mostly to employment and regional disparities. Other topics related to citizens, questions about ethical impacts of technology and innovation, and wider issues of citizens' interactions with STI are not yet among the core priorities of strategies. The health topic is directly a reflection of the COVID-19 pandemic that led to a focus on the question within STI strategies aside from strategies on health.

Fifth, new approaches to STI policy reflected in strategies include joint governance (5.4%) - comprising vertical governance and consultations-, policy agility (5.1%) -with emphasis on monitoring and evaluation and foresight-, as well as directional policy (2.3%). This is noteworthy and widespread across countries, indicating awareness that new goals set need different approaches to implementing STI policy. A closer look at specific examples illustrates a rich diversity of approaches implemented by STI policy makers.

Sixth, a robustness analysis of label accuracy across topic variations highlights the consistency and variability in topic representation within the strategies. Innovation Policy 3.0 is the most represented category, yet the analysis reveals that the 35-label model (the baseline) slightly underrepresents its prevalence compared to alternative models. Traditional innovation policies and new innovation policy approaches vary significantly based on the R&D spending level of the countries, with lower R&D countries focusing more on traditional approaches. The inclusion of miscellaneous topics, despite their procedural nature, shows significant representation across the corpus, particularly when high-impact countries like the United States are excluded.

Seventh, methodologically the paper shows that topic modelling can provide valuable comparative insights into the policy goals countries discuss in their STI strategies. The strength of this analysis is that it enables the identification of dominant discussions and themes across a large volume of documents, reducing biases associated with selective reading or purely qualitative approaches. However, topic modelling also necessitates extensive document preprocessing, such as the removal of stop words, and requires programming skills. Moreover, it relies on human decisions that may introduce subjectivity, particularly in selecting the number of topics and interpreting or labelling the results. Additionally, there is a risk of overlooking important details and nuances in policies and programmes that may not appear in the final topic model.

The work relates to two strands of literature. The first is the extensive literature on Innovation Policy 3.0, including notably several empirical studies. This has been articulated in several contributions by Haddad and Bergek (2023^[7]), Edler and Boon, (2018^[8]), Schlaile et al., (2017^[9]), Weber and Rohrer (2012^[10]) and Grillitsch et al. (2019^[3]). These STI policy evaluations rely on qualitative methods such as interviews and qualitative document analysis, complemented by using available quantitative indicators. The second strand connects to the quickly expanding literature that applies NLP, which is widely used in adjacent fields such as in economics (Gentzkow, Kelly and Taddy, 2019^[11]) or political sciences (Benoit, 2020^[12]), but has so far had limited application in innovation policy. In innovation-related works, NLP has also been applied widely to patent and publications data analysis (e.g., Arts et al. (2021^[13])).

The remainder of this article proceeds as follows. Section 2 provides a conceptual framework. Section 3 presents the empirical strategy, the database used, and some methodological considerations. Section 4 presents the findings. Section 5 concludes.

2 Conceptual framework

According to Schot and Steinmueller (2016^[14]), the development of innovation policy can be understood through the lens of three successive "framings" (Grillitsch et al., 2019^[3]). Each framing relates to theoretical models that differ in the STI policy goals to be achieved, their underlying rationale and consequently the tools required.

"Innovation Policy 1.0" emerged in the 1960s and was strongly influenced by neo-classical welfare economics and growth theories. The model of innovation in Innovation Policy 1.0 is focused on the discovery stage because the market may underinvest in such growth-enhancing research (Laranja, Uyarra and Flanagan, 2008^[15]). Policy interventions in the innovation process are then considered justifiable where markets would not invest in basic research due to their inability to privately appropriate the wider societal gains from this endeavour (Howlett, Ramesh and Perl, 2020^[16]). The main tool applied here is the provision of research funding conducted by research institutions and firms.

During the 1980s, the second framing of innovation policy, "Innovation Policy 2.0," increasingly focused on *systems* of innovation, a "horizontal policy approach that mobilises technology, market mechanisms, regulations and social innovations" (OECD, 2015, p. 7^[17]). The rationale for policy interventions shifted to system failures that emerge from missing or malfunctioning linkages between research institutions and firms. New instruments consequently focus on boosting industry-science linkages and knowledge transfer.

The latest framing of STI policies – Innovation Policy 3.0 – sets societal goals at the heart of STI policy (Diercks, Larsen and Steward, 2019^[4]; Asheim, Moodysson and Tödtling, 2011^[5]; Mazzucato, 2016^[6]), beyond specific achievements within the STI field as was the case of Innovation Policy 1.0 (boosting basic research) and Innovation Policy 2.0 (boosting industry-science linkages). As Schot and Steinmueller (2016, p. 17^[14]) put it, "innovation policy should focus much less on products, processes, firms, and R&D, but on the achievement of systems wide transformations." The emphasis on "failures" inherent to the first two framings is replaced by more pro-active, purposive, and developmental approaches to innovation policies (Asheim, Moodysson and Tödtling, 2011^[5]; Mazzucato, 2016^[6]). Here, the model of innovation emphasises socio-technical systems and how processes of innovation and technological change are embedded in society. Innovations are not seen as beneficial per se but are judged by their positive or negative externalities for society.

In this Innovation Policy 3.0 period, new policy imperatives for policy become essential (Arnold et al., 2023^[18]), which are: (i) joint governance, referring to both effective coordination across different levels of government but also stakeholder engagement. Joint governance is needed as reaching ambitious societal transformation goals needs coordinated government efforts from local to national levels and working with diverse actors in industry, research, government, and society. Other imperatives are (ii) building more agile and experimental policies, and (iii) setting policy targets to reach societal goals (directed policy). Directionality is important as coordinated action to advance STI in specific areas, particularly in green innovation, is otherwise not necessarily forthcoming. Furthermore, (iv) policy agility and experimentation are also needed as transitions are complex and uncertain processes that require adaptive and flexible approaches. Policy experimentation allows for testing and learning from new ideas and interventions, essential in a new context with impacts on policies' success. This generation of policies owes its

emergence to the twin transition debates marked by the changes introduced by digital technologies and the green transition imperative.

In conclusion, an assessment of Innovation Policy 3.0 would start by identifying the topics specified in countries' STI strategies. These would be societal goals that are ambitious in nature as they look for transformations, notably as regards building green futures and shaping digitalisation in societally desirable ways. They differ from previous generations of policies that were more inward-focused, relating to business and research institutions and their performance, focused on productivity. They would also discuss new approaches to policy making.

3 Data and empirical strategy

To assess to what extent the new frame of “Innovation Policy 3.0” has been adopted, the paper examines national STI strategies of 24 OECD countries.

Why strategies?

The analysis focuses on official STI policy strategies issued by national government entities to evaluate whether innovation policies shifted to the Innovation Policy 3.0 paradigm. It follows Wauters (2019^[19]) in defining strategies as: *“coherent action, guided by a longer-term sense of direction, to transform value-generating processes, undertaken by (a network of) actors, backed by an argument.”* In this definition: (1) strategy is inextricably related to action; (2) action must be coherent with respect to a long-term objective; (3) strategy is related to organising the foundational questions of what government should do (referred to as “value-generating processes”) and not to managing the minutiae of regular activity, which Ruegg-Sturm (2005^[20]) distinguishes as mere “tactics”. Strategy is backed by an argument in favour of how and why it occurs. The argument for “why” suggests that a strategy requires an articulation of why the issues it targets are important. The argument for “how” relate to what Weiser et al. (2020^[21]) call a “theory of change”, which is an understanding of how the actions in a strategy cause the desired outcomes.

Consequently, STI strategy documents were selected for the assessment of Innovation Policy 3.0 as they capture governments’ value-embedded, forward-looking, and high-level goals and plans. Moreover, to the extent that policy making is responsive to citizens, STI strategies should also partly reflect societal endorsement of the directions of STI policy.

Database of STI strategy documents

A total of 313 documents were gathered, representing 24 OECD countries’ STI strategies up until 2021, thus covering the beginning of the COVID-19 pandemic. Strategies were identified based on their contents presenting forward-looking goals for future policy actions and priorities. To understand common trends, the country selection represents the variety of OECD countries across Europe and beyond. In Europe, it covers Northern Europe (Finland, Ireland, Sweden, United Kingdom), Western Europe (Austria, Belgium, Germany, Luxembourg, Netherlands, Switzerland), Eastern Europe (Hungary, Slovakia) and Southern Europe (Greece, Italy, Spain). It also covers North and South America (Canada, Chile, Costa Rica, United States) as well as Asia-Pacific (Australia, Korea, Japan, New Zealand). Within those regions, the selection was also based on including diverse countries, specifically regarding their sizes.

Following an initial identification of documents based on the EC-OECD STIP Compass database as well as other sources, policy makers from the respective governments validated the selection, identified documents that were erroneously included, and provided additional relevant documents. The approved document list was then used in the analysis.

The final database is summarised in Table 1 (with the full list of documents provided in the Annex Table 3. List of documents analysed). The diversity in terms of total word and document counts across countries is substantial, likely due to different ways of issuing and drafting strategies. Also, the time period is shorter for countries that frequently issue new strategies replacing old ones and much longer for others.

Table 1. Descriptive statistics for the final database of STI policy strategy documents

Country	Time period	Number of documents	Share in total (documents)	Number of words original	Number of words in final corpus	Share in total (words final)
Australia	2015-2021	14	4.5%	140,320	50,691	1.5%
Austria	2016-2021	10	3.2%	134,507	45,095	1.3%
Belgium	2017-2021	7	2.2%	633,403	194,508	5.6%
Canada	2014-2021	21	6.7%	754,915	221,005	6.4%
Chile	2015-2021	13	4.2%	343,716	103,955	3.0%
Costa Rica	2014-2021	16	5.1%	248,469	72,048	2.1%
Finland	2017-2021	8	2.6%	286,677	120,943	3.5%
France	2013-2021	11	3.5%	583,469	168,042	4.8%
Germany	2015-2021	24	7.7%	643,060	236,762	6.8%
Greece	2014-2021	9	2.9%	674,231	213,209	6.1%
Hungary	2014-2021	16	5.1%	584,171	231,670	6.7%
Ireland	2015-2021	15	4.8%	258,242	78,542	2.3%
Italy	2013-2021	11	3.5%	440,743	152,858	4.4%
Japan	1995-2021	25	8.0%	345,994	378,435	10.9%
Korea	2019-2021	7	2.2%	93,839	46,554	1.3%
Luxembourg	2019-2021	9	2.9%	298,179	92,928	2.7%
Netherlands	2014-2021	11	3.5%	233,128	74,559	2.1%
New Zealand	2015-2020	8	2.6%	260,144	77,514	2.2%
Slovakia	2013-2021	11	3.5%	713,623	267,996	7.7%
Spain	2015-2021	16	5.1%	689,949	193,046	5.6%
Sweden	2016-2021	10	3.2%	335,173	122,493	3.5%
Switzerland	2016-2021	8	2.6%	141,801	38,205	1.1%
United Kingdom	2019-2021	9	2.9%	152,876	48,046	1.4%
United States	2017-2021	24	7.7%	762,962	245,518	7.1%
Total		313	100%	9,753,591	3,474,622	100%

Source: Authors' own elaboration.

Moreover, the types of documents differ and include national STI strategies, sector-specific strategies that explicitly refer to STI, strategies of agencies or ministries in charge of STI, legislation on STI policy for countries where legislative documents shape STI objectives and, for EU countries, the COVID-19 recovery packages and legislative measures given their strong focus on STI.

To prepare the database, several pre-processing and cleaning steps were applied (see Annex for a full description), resulting in 313 text files, each containing the central semantic content but far fewer words than the original documents (see Table 1 above). These text files form the final corpus used in the subsequent analysis.

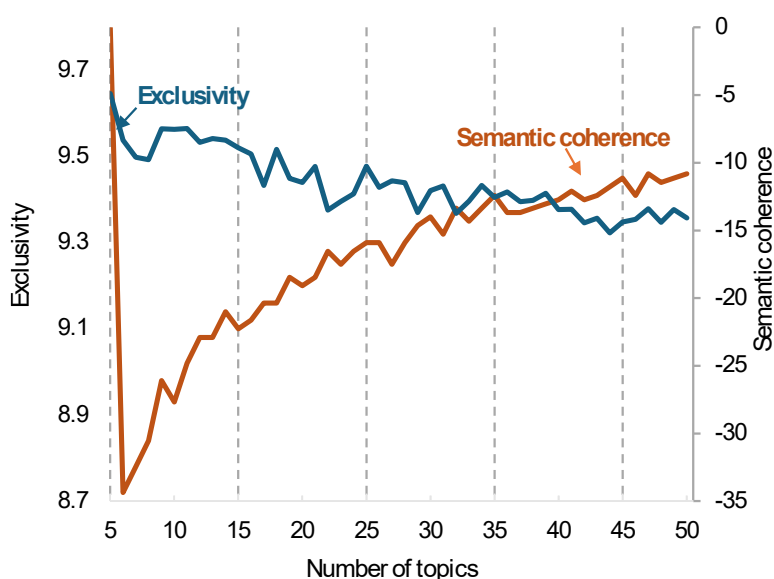
Topic modelling and extraction of relevant text

For the analysis of the pre-proceed and text files, topic modelling is employed - a natural language processing technique that uncovers “topics” within large amounts of textual data (Blei, 2012^[22]; Hannigan et al., 2019^[23]). Topic modelling is an unsupervised machine learning tool, which means that it does not require prior data inputs such as hand-labelled text passages but extracts the different topics solely from the given textual data. This makes it useful for the analysis of our large and heterogenous corpus that cannot be easily and consistently analysed through qualitative reading of each individual document.

Specifically, Structural Topic Modelling (STM) is employed, which is an extension of the often-used Latent Dirichlet Allocation (LDA) algorithm. The basic idea behind LDA is to treat documents as a mixture of latent topics and words as a distribution of these topics, enabling it to represent each document with a probability distribution over topics and each topic with a probability distribution over words (Blei, 2012^[22]). By iteratively adjusting these probabilities to optimally capture the different topics in a text, LDA identifies the best-fitting topics and their corresponding word distributions, generating an overview of documents’ semantic content. STM extends LDA to incorporate non-textual information for each document to better capture semantic nuances in the textual data and enhance the interpretability of topics. STM is particularly useful when analysing data with inherent grouping, such as by country and year of publication of the different documents in our database. Technically, the STM models the relationship between topic proportions and covariates (country and year of publication) through a hierarchical Bayesian framework, and considers how document-level covariates influence these proportions in the process of identifying topics (Roberts, Stewart and Tingley, 2019^[24]).

The optimal number of topics for the corpus was determined based on two statistical measures of exclusivity and semantic coherence (Figure 1), and an assessment of the clarity of the topics obtained. Exclusivity measures the extent to which the most frequent words in one topic are infrequent in other topics as developed by Roberts et al. (2014^[25]). As shown in Figure 1, exclusivity decreases with the number of topics selected in the corpus. Semantic coherence assesses the extent to which the most frequent words in a topic co-occur with one another, as proposed by Mimno et al. (2011^[26]). The measure evolves reversely with the number of topics, since frequent words co-occurrence decreases (increases) with more (fewer) topics, though not fully linearly. A total of 35 topics were selected, since semantic coherence increases substantially over those initial topics but less substantially afterwards while exclusivity does not decrease substantially. The number was also suitable in providing clarity of distinct topics.

Figure 1. Exclusivity and semantic coherence for distinct numbers of topics



Source: Authors' own elaboration.

Topics were subsequently labelled and grouped into overarching topic areas. For each topic, the list of most common words as well as the words that are both frequent and unique for each topic were extracted – as measured by the FREX metric (Bischof and Airoldi, 2021^[27]). FREX weights the frequency of a word occurring in a given topic by its exclusivity in order to identify words that are most “representative” of a certain topic (see both lists in the tables below). Using the words with the highest FREX value, labels were assigned to capture the semantic content of each of the 35 topics. These topics were then grouped into broader themes and categories, as shown in Annex Table , with corresponding coverage in the corpus as shown in

Table 2.

For the analysis, text fragments classified by the topic model as being highly representative of a specific theme were extracted based on the maximum a posteriori (MAP) approach. This approach estimates how many words in a short fragment of text are assigned to certain topics. This method was applied to enable a more in-depth exploration of new approaches contained in strategies on joint governance, policy agility and directional policy. The MAP approach is a useful addition to qualitative analyses because it allows to specifically target a certain topic among all 313 STI strategy documents included in the database, without reading the entirety of documents or searching for single key words of interest.

Table 2. Shares of coverage of all topics, themes and categories in the corpus

Nb.	Topic	of 35	% within theme	within category	Theme	of 35	% within category	Category	% of 35
1	COVID-19 recovery and resilience	2.5%	100.0%	51.5%	COVID-19 and resilience	2.5%	51.5%	Context	4.9%
2	Global energy prices	2.4%	100.0%	48.5%	Energy prices	2.4%	48.5%		
3	Growth	4.0%	45.4%	21.8%	Supporting firms, industry and growth	10.8%	58.3%	Traditional innovation policy	18.6%
4	Investment and productivity	2.6%	28.6%	13.7%					
5	Conditions for growth	2.3%	26.0%	12.4%					
6	Firms	1.9%	31.1%	10.3%					
7	Research	4.3%	55.6%	23.1%	Supporting research and industry-science linkages	7.7%	41.7%		
8	Support for industry-science linkages	3.5%	45.4%	18.9%					
9	Energy CO2 emissions	3.5%	18.7%	8.3%	Green transition goals	18.6%	44.5%	Innovation policy 3.0	41.7%
10	Power generation	2.3%	12.5%	5.6%					
11	Energy efficiency	2.1%	11.2%	5.0%					
12	Sustainable industries	3.2%	17.3%	7.7%					
13	Clean mobility	3.0%	16.3%	7.3%					
14	Pollution mitigation	2.2%	12.0%	5.4%					
15	Climate change and adaptation policy	2.2%	11.9%	5.3%					
16	Capacity building	4.1%	40.9%	9.7%	Social and regional inclusion	9.9%	23.7%		
17	Inclusion in the labour market	3.9%	39.1%	9.3%					
18	Regional development	2.0%	19.8%	4.7%					
19	Technologies for Industry 4.0 and defence	3.5%	36.9%	8.4%	Digital and technological transformation goals	9.5%	22.7%		
20	Digital infrastructure	3.0%	31.5%	7.1%					
21	Data	3.0%	31.7%	7.2%					
22	Health, knowledge and technology	3.8%	100.0%	9.1%	Health knowledge and technology	3.8%	9.1%		
23	Consultation	3.4%	64.1%	26.9%	Joint governance	5.4%	41.8%	New innovation policy approaches	12.8%
24	Vertical governance	1.9%	35.4%	14.9%					
25	Monitoring and evaluation	2.6%	52.0%	20.6%	Policy agility	5.1%	39.9%		
26	Foresight	2.5%	48.7%	19.3%					
27	Directional policy	2.3%	100.0%	18.3%	Directional policy	2.3%	18.3%		
28	Design	2.5%	22.9%	11.2%	Strategy and procedural aspects	10.8%	48.9%	Misc. topics	22.0%
29	Strategy	3.2%	29.6%	14.5%					
30	Documents	3.3%	30.6%	15.0%					
31	Location and time context	1.8%	16.2%	8.0%					
32	Policy intervention	2.7%	23.8%	12.2%	Government action	11.3%	51.1%		
33	Tax	3.9%	34.9%	17.9%					
34	Regulatory and legal reform	2.4%	10.9%	10.7%					
35	Regulation	2.3%	20.1%	10.3%					

Source: Authors' own elaboration.

Robustness analysis of label accuracy across topic variations

In addition to the statistical analysis of exclusivity and semantic coherence for identifying the optimal number of topics, a further robustness test was conducted to assess the accuracy of the results and the selected baseline of 35 topics (see Figure 3). This test examined how the percentages varied across different labels, considering both lower numbers of topics (25 and 30) and higher numbers (40 and 45).

To assess the impact of specific countries, the core five identified countries—Canada, Germany, Japan, Korea, and the United States—were each excluded separately from the sample during these checks. This approach allowed for an analysis of how the inclusion or exclusion of each country affected the overall percentages allocated to the remaining 23 OECD countries.

Strategic variations in STI policy were also explored under three different scenarios: the top 10 countries with the lowest R&D spending in the sample, the top 10 countries with the highest R&D spending, and topics based solely on specific STI strategies.

These variations involved reviewing the percentage allocations for each topic across different topic numbers and comparing them with the 35-topic baseline. The goal was to identify any discrepancies or variations, such as the underrepresentation of certain topics, and to assess how these variations impacted the model.

The method for generating the labels remained consistent with the approach used to create the baseline model of 35 topics, ensuring comparability across all tests and scenarios.

Methodological considerations

The main advantage of topic modelling lies in generating high-level summaries of documents' contents by identifying clusters of co-occurring words that represent key themes in the text. This saves time and avoids the biases of human reading and structuring of textual information.

However, topic modelling also has its limitations. This includes:

- The need for extensive document preprocessing, such as the removal of stop words, and programming skills.
- Reliance on human decisions that may introduce subjectivity, particularly in selecting the number of topics and interpreting or labelling the results.
- Risk of overlooking important details and nuances in policies and programmes that may not appear in the final topic model.
- Risk of an incomplete or unbalanced document corpus, despite validation efforts, where key texts might be missing or irrelevant ones included, potentially limiting the representativeness and significance of the policy debate captured.

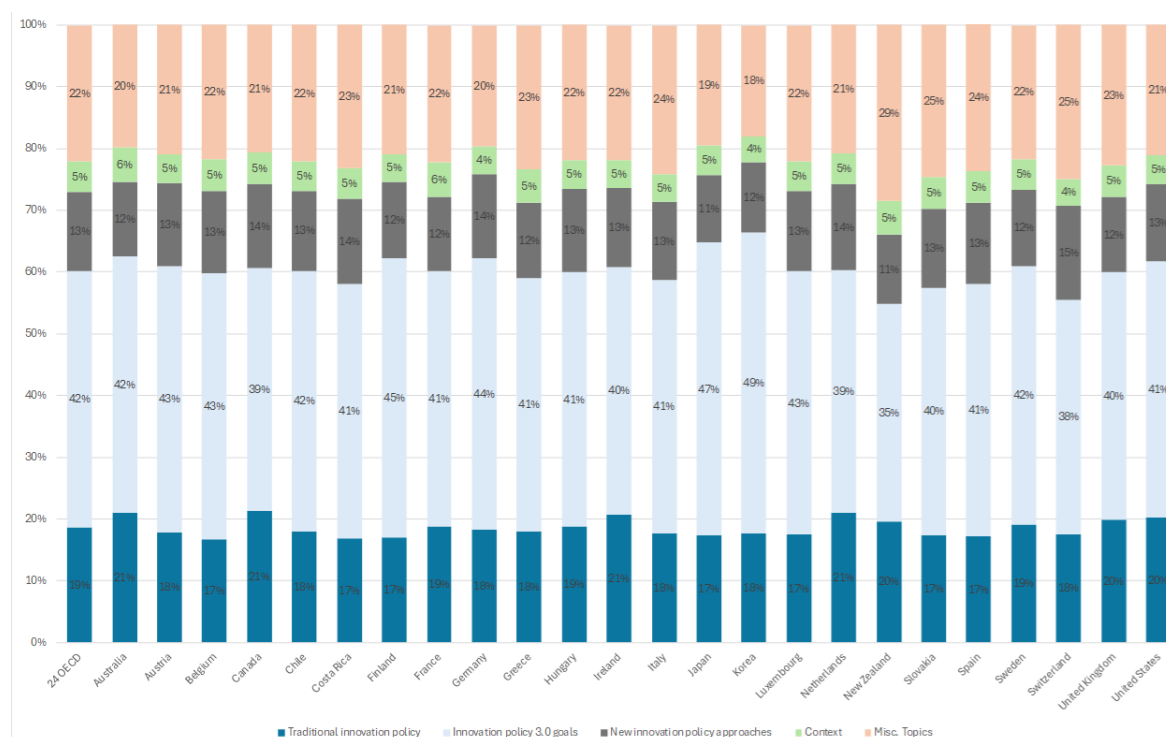
Additionally, while topic modelling can process vast amounts of data and avoid the subjectivity of qualitative reading, it lacks the interpretative capacities and knowledge required to assess the rigorousness and meanings of as well as the relations between information extracted from documents. Validation of all analytical outputs by domain experts is thus necessary to ensure the accuracy and proper contextualisation of outputs generated through computational tools.

4 Findings

Distribution of discussions in strategy documents

Looking into the overall distribution of topic categories, goal setting dominates strategy documents over discussions on new policy approaches to reach those goals and on the socio-economic context to which strategies respond. STI policy objectives – including both those that are traditional and Innovation Policy 3.0 – represent on average 60.3% of the substantive topics in the corpus of strategies in 24 OECD member countries (24 OECD). The combined share of STI policy objectives range from 54.8% to 66.3% across countries, driven mainly by stronger (weaker) emphasis on new STI policy approaches where the share is lower (higher) (Figure 2).

Figure 2. Shares of topic categories across strategies in 24 OECD member countries



Source: Authors' own elaboration.

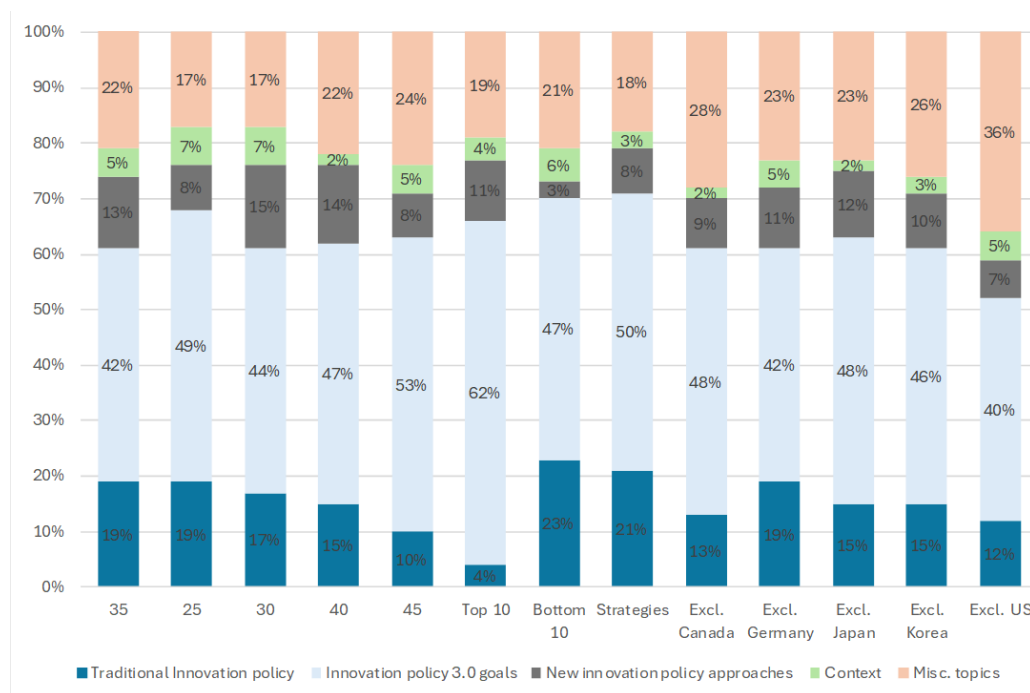
Another substantive topic category relates to new STI policy approaches, representing 12.8% of the total topics modelled. While possibly also discussed in other official documents, the lower share suggests that

goal setting is not immediately connected to implementation issues. The goals set out in strategies are consequently not necessarily reflected in new ways of implementing innovation policies. Nonetheless, the reference in all countries' corpora to new approaches – as discussed further below – indicates awareness of the policy community that sets goals that new ways of doing STI policy need to be embraced.

Finally, 22% of topics in documents contain references to procedural dimensions of strategies (design, context, etc.) and government actions (tax collection, regulations, etc.) and 4.9% to the for policies (4.9% on average), covering mostly introductory sections of strategies. Unsurprisingly, given the period of the analysis, these context topics are COVID-19 and building resilience to shocks, including disturbances to international supply chains (Bas, Fernandes and Paunov, 2023^[28]) and the energy crisis. These context variables confirm the timely nature of STI strategies intent on responding to societal challenges, rather than treating structural features of innovation policies as if they existed in isolation from the broader context.

Sensitivity tests—varying the number of topics, focusing on top and bottom R&D performers, using strategies only, and excluding core countries—highlight the consistency of topic categories. Interestingly, the share of Industry 3.0 goals – that is 20 percentage points more important among the top 10 R&D innovating countries compared to the baseline, while only 4% of their corpus relate to traditional innovation policy topics. Miscellaneous topics (e.g. procedural and governance-related themes) are more important when the US data are excluded as compared to the baseline and all other alternative models (35 topics, the average for 24 OECD countries; see Figure 3).

Figure 3. Share of topics categories across model variations



Source: Authors' own elaboration.

Note: The figure shows the percentage of the corpus attributed to Innovation Policy 3.0 under different variations. These include changes in the number of topics (from the baseline of 35, adjusted to 30, 25, 40, or 45 topics), analyses limited to the top 10 and bottom 10 R&D-spending countries (TOP 10 and LOW 10), and the exclusion of core countries (Canada, Germany, Japan, Korea, and the United States).

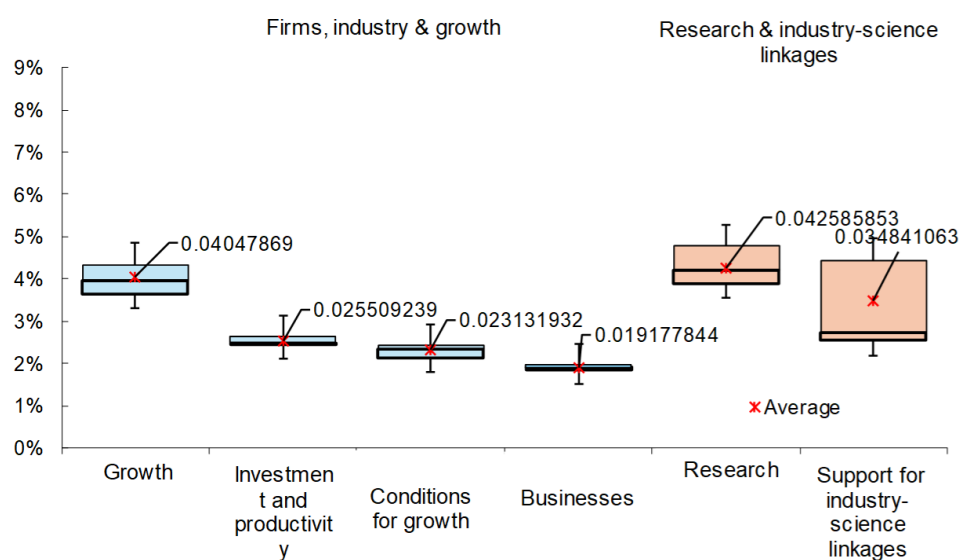
Traditional and innovation policy 3.0 goals

Comparing topics related to goals of Innovation Policy 3.0 with traditional innovation policy, that the analysis finds that these are more than twice as important as traditional innovation policy topics (41.7% versus 18.6% of the total topics modelled). This applies to all countries, with a range of the Innovation Policy 3.0 share of 13.5 percentage points across the 24 countries (Figure 2). This confirms that 24-OECD countries' strategies have embraced new goals for innovation policy.

Nonetheless, when varying the sample and considering only those countries with lower levels of R&D spending, the representation that traditional innovation gets over the medium is quite significant (around six percentage points more – a total of 23.5%). In accordance, the levels of allocation of the top R&D spending countries on traditional policies are lower (fifteen percentage points less than the mean – 3.4%). This indicates that countries that have lower levels of R&D spending typically focus more on traditional approaches, whereas the ones that have higher levels of spending are more focused on allocating their resources towards innovation 3.0 and new innovation policy approaches.

Traditional innovation policy concerns – relating to industry and research actors and their connection – make up 18.6% of the total topics modelled. Within the firms, industry and growth theme, the topics are businesses (1.9%), growth (4%), conditions for growth (2.3%) and investment and productivity considerations (2.6%) (Figure 4). On the side of research, support for industry-science linkages (3.5%) is emphasised, following the recognition of connections that emerged from the national innovation systems framework (Nelson, 1993^[29]; OECD, 1997^[30]; Lundvall, 2007^[31]). Variations across countries are small across most topics with the largest variation in industry-science linkages – with a range of 4.4% – and to a lesser extent research and growth – with ranges of 2.9% and 2.1%, respectively.

Figure 4. Cross-country distribution of the shares of traditional innovation policy goals



Source: Authors' own elaboration.

Note: The rectangles indicate the median, 75th percentile, and 25th percentile of topic shares among the 24 country corpuses. The upper and lower lines represent the maximum and minimum values.

Robustness tests confirm the strong presence of Innovation Policy 3.0 across a range of model variation, with a mean of 47.4% across all variations (see Table 3). These include changes to the number of topics (25, 30, 40, and 45, compared to the baseline of 35), the exclusion of core countries (Canada, Germany, Japan, Korea, and the United States), and separate analyses of strategies from the top 10 and bottom 10 R&D spenders. In fact, the share of Innovation Policy 3.0 is lowest at the baseline of 35 topics and increases with all other topic numbers. The share allocated to Innovation Policy 3.0 ranges from 40.3% (when core countries are excluded) to 61.5% (when analysing STI strategies only). When considering only topic-number changes, the mean is 46.8%.

Table 3. Share of Innovation policy 3.0 across topic number and sample variation

Number of topics					Corpus selection							
25	30	35	40	45	Top 10	Bottom 10	Strategies	Excl. Canada	Excl. Germany	Excl. Japan	Excl. Korea	Excl. US
48.6	44.4	41.7	46.8	52.6	61.5	47.0	50.0	47.8	42.1	47.6	46.0	40.4

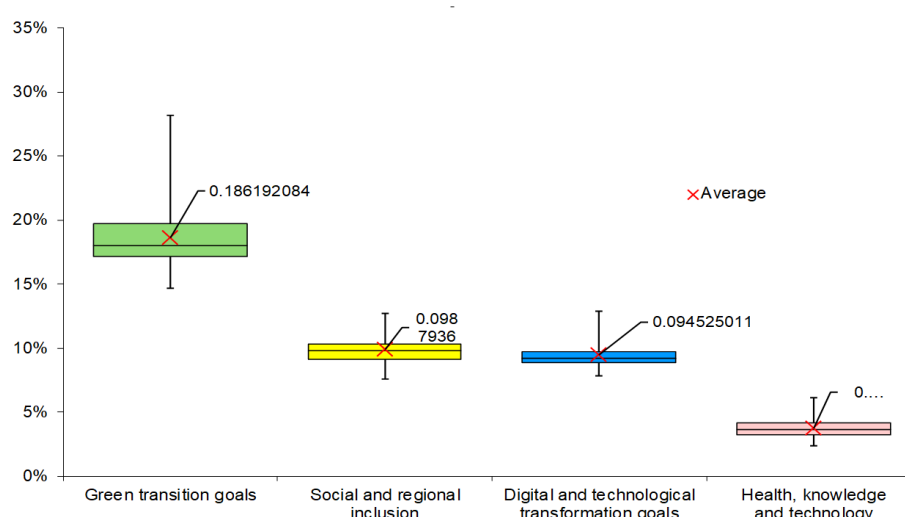
Source: Authors' own elaboration.

Note: The table shows the percentage of the corpus attributed to Innovation Policy 3.0 under different variations. These include changes in the number of topics (from the baseline of 35, adjusted to 30, 25, 40, or 45 topics), analyses limited to the top 10 and bottom 10 R&D-spending countries (Top 10 and Bottom 10), and the exclusion of core countries (Canada, Germany, Japan, Korea, and the United States).

As to the Innovation Policy 3.0 category, green transition goals (18.6%) dominate, with nearly double the importance of other themes and 7 of the 14 topics in this category (Figure 5). Energy generation sources, CO2 emissions, and efficiency are a predominant theme, accounting for 7.9% of the corpus. Sustainable industries (3.2%) and clean mobility (3%) are also targets (Table 5).

Other green strategy topics regard pollution mitigation and climate change and adaptation policies. Absent from key themes are biodiversity, circular economy, and climate resilience. The lower share of the latter themes is likely a result of a lower prioritisation of certain themes in policy debates on climate change and the (perceived) lower contributions of STI towards some of these goals.

Figure 5. Cross-country distribution of the shares of innovation policy 3.0 goals



Source: Authors' own elaboration.

Note: The rectangles indicate the median, 75th percentile, and 25th percentile of themes' shares among the 24 country corporuses. The upper and lower lines represent the maximum and minimum values.

There are, however, important differences across countries in the emphasis they set on specific green transition topics. For instance, Belgian strategies emphasise green mobility (4.3% of the total topics modelled) well above the 24-OECD countries average (3%), while Japan – because of the Fukushima nuclear accident of 2011 – is strongly focusing on power generation (7.6%). This extensive focus also results in Japan setting more emphasis on the topic of green transitions compared to other countries. This underscores that direct country experiences expedite policy action for the green transition as compared to policy actions to mitigate anticipated future changes.

Sensitivity tests confirm the robustness of green transition goals, with some variation observed only when Japan is excluded (see Table 4). The importance of green goals remains stable across different topic selections and among both top and bottom R&D performers. Interestingly, green goals are about 4 percentage points higher among top innovators as compared to others. Yet, green transition goals drop to just 14.2% when Japan is excluded—highlighting Japan’s outsized role in driving the prominence of green transition objectives (The exclusion of Germany and Canada reduces the share of green topics by 1.3-1.4 percentage points compared to the baseline).

Table 4. Share of green transition goals across topic number and sample variation

Number of topics					Corpus selection							
25	30	35	40	45	Top 10	Bottom 10	Strategies	Excl. Canada	Excl. Germany	Excl. Japan	Excl. Korea	Excl. US
27.6	28.1	18.6	19.2	19.5	23.9	19.5	20.3	17.2	17.3	14.2	22.2	19.4

Source: Authors’ own elaboration.

Note: The table shows the percentage of the corpus attributed to green transition goals under different variations. These include changes in the number of topics (from the baseline of 35, adjusted to 30, 25, 40, or 45 topics), analyses limited to the top 10 and bottom 10 R&D-spending countries (TOP 10 and LOW 10), and the exclusion of core countries (Canada, Germany, Japan, Korea, and the United States).

Social and regional inclusion topics constitute an important 9.9% of the total topics modelled. This theme primarily revolves around labour market issues, focusing on capacity-building endeavours to enhance skills (4.1%). It also addresses the challenges of gender, disability, and poverty, reflecting a broader commitment to labour market inclusion (3.9%). The analysis also delves into the local and regional dimensions of STI policies, crucial conduits that can either foster or hinder inclusion, accounting for 2% of the discussion (Table 5).

In addition, following with further check on reference labels and sample variations, when examining only the top ten R&D spenders, the percentage of representation of social and regional inclusion strategies raises sharply – to 22.6%. This indicates that countries that invest a lot on R&D are not only prioritising technological aspects but also placing importance on social concerns.

Equally significant is the 'digital and technological transformation goals' theme, comprising 9.5% of the total topics modelled. This theme addresses a range of issues, from the infrastructure needed for digital transformation to the use of public data, especially in healthcare, and emerging technologies like quantum and AI. The lower emphasis in STI strategies on the digital theme, compared to green transitions, is partly because governmental responsibilities are shared, including on regulatory matters related to competition policy and consumer protection. Green transitions also involve many government entities, but the immediate need to foster green STI to support green transitions contributes to their prominence in STI strategies. This differs from digital technologies where swift progress requires extensive action on other sides, such as regulations and infrastructures, to advance on this transition.

The final theme within the Innovation Policy 3.0 goals agenda refers to health, knowledge, and technology (3.8%). The health topic is directly a reflection of the COVID-19 pandemic that led to a focus on the question within STI strategies aside from strategies on health.

Table 5. Distribution of shares of innovation policy 3.0 topics across countries

Theme	Topic	Average	Median	p25	p75	Min.	Max.	Max. (Country)
Green transition goals		18.6%	18.0%	17.1%	19.8%	15.1%	27.1%	Japan
	Climate change and adaption policy	2.2%	2.2%	2.0%	2.4%	1.8%	2.9%	Slovakia
	Pollution mitigation	2.2%	2.3%	2.1%	2.4%	1.9%	2.7%	Finland
	Power generation	2.3%	2.2%	1.9%	2.3%	1.5%	7.6%	Japan
	Energy efficiency	2.1%	1.8%	1.6%	1.9%	1.3%	6.0%	Korea
	Energy CO2 emissions	3.5%	3.3%	3.1%	4.0%	2.3%	5.1%	Belgium
	Clean mobility	3.0%	2.9%	2.6%	3.3%	2.2%	4.3%	Belgium
	Sustainable industries	3.2%	3.3%	3.0%	3.5%	2.2%	3.9%	Costa Rica
Social and regional inclusion		9.9%	9.8%	9.2%	10.3%	8.3%	12.5%	Finland
	Inclusion in the labour market	3.9%	3.9%	3.5%	4.2%	3.0%	5.0%	Finland
	Capacity-building	4.1%	4.1%	3.7%	4.4%	2.9%	5.3%	Finland
	Regional development	2.0%	1.9%	1.8%	2.1%	1.7%	2.3%	Greece
Digital and technological transformation goals		9.5%	9.2%	8.9%	9.7%	7.8%	12.4%	United States
	Technologies for Industry 4.0 and defense	3.5%	3.3%	2.9%	3.7%	2.5%	6.7%	United States
	Digital infrastructure	3.0%	3.0%	2.8%	3.2%	2.3%	3.6%	Greece
	Data	3.0%	2.9%	2.8%	3.1%	2.6%	3.7%	Korea
Health, knowledge and technology		3.8%	3.7%	3.3%	4.2%	2.6%	5.7%	Chile
	Health, knowledge and technology	3.8%	3.7%	3.3%	4.2%	2.6%	5.7%	Chile

Source: Authors' own elaboration.

New approaches to reaching “Innovation 3.0” goals

Aside from goals, new STI policy approaches for reaching Innovation Policy 3.0 goals reflected in the 35 topics identified are: i) joint governance (5.4%), ii) policy agility (5.1%) and iii) directional policy (2.3%) (see Table 6). These topics cover jointly 10.6% of the substantive topics of the strategies. The coverage illustrates awareness across countries about the need for new policy approaches so that societal goals can be reached.

Table 6. New innovation policy approaches

Theme	Topic	Average	Median	p25	p75	Min.	Max.	Max. (Country)
Joint governance		5.4%	5.3%	4.9%	5.6%	4.6%	7.0%	Canada
	Consultation	3.4%	3.4%	3.1%	3.6%	2.8%	5.2%	Switzerland
	Vertical governance	1.9%	1.8%	1.6%	2.0%	1.4%	3.6%	Canada
Policy agility		5.1%	4.9%	4.5%	5.6%	4.2%	6.3%	Slovakia
	Monitoring and evaluation	2.6%	2.7%	2.2%	2.9%	2.0%	3.8%	Costa Rica
	Foresight	2.5%	2.3%	2.2%	2.7%	2.1%	3.4%	Slovakia
Directional policy		2.3%	2.2%	2.1%	2.5%	1.8%	3.3%	Germany
	Directional policy	2.3%	2.2%	2.1%	2.5%	1.8%	3.3%	Germany

Source: Authors' own elaboration.

Regarding joint governance, the 35 topics modelled contain two related topics i) vertical governance (1.9%), i.e. the linkages between national and regional policies, which – as outlined in our conceptual framework above – is a key policy imperative to operate in transitions. Unsurprisingly, Canada, a federal country where states play important roles in innovation policies features, stands out with an even higher share of coverage (of 3.6%). Table 7 provides a few specific examples from the documents on this theme, such as an inter-ministerial coordination group on green growth in Finland (a. in Table 7) and municipal real laboratories for energy transitions in Germany (b. in Table 7).

Table 7. Examples of new innovation policy approaches from the corpus

	Issue discussed	Country	Year	Title	Institution
	Joint governance				
a.	Inter-ministerial coordination group on green growth	Finland	2021	Finland's Sustainable Growth Agenda: Preliminary Recovery and Reconstruction Plan	Ministry of Finance
b.	Municipal real laboratories for Germany energy transition experiments	Germany	2021	Germany Recovery and Resilience Plan	Federal Ministry of Finance
c.	Public engagement on nuclear power risks and uses	Japan	2021	Energy Situation and Main Measures	Ministry of Economy, Trade and Industry
d.	Smart cities collaboration	Sweden	2016	Government Research Bill	Government of Sweden
	Agility				
e.	Development of future scenarios for policy for cities exposed to natural threats	Chile	2012	Report Towards a Disaster Resilient Chile: An Opportunity	National Council for Innovation for Development
f.	Support for technology foresight activities of local decision-makers	Italy	2020	Horizon 2002	Ministry of Education, University and Research
g.	Establishment of a long-term evaluation and monitoring system to measure the outputs and impacts of Hungary's climate action plan	Hungary	2020	Climate Change Action Plan	Ministry of Innovation and Technology
h.	Establishment of local climate roundtables and use of their data for monitoring Belgium's energy and climate plan	Belgium	2019	National Energy-Climate Plan	National Climate Commission
	Directional policy				
i.	Support for sustainable transportation modes in cities	Costa Rica	2015	Actional Plan for Science, Technology and Innovation 2015-2021Ministry of Science, Technology and Telecommunication	Ministry of Science, Technology and Telecommunication
j.	Energy performance diagnosis regulations for housing	France	2021	National Recovery and Resilience Plan	Governemtn of France
k.	Green public procurement in the housing sector	Korea	2020	2050 Carbon Neutral Strategy	Government of Korea
l.	Investment in alternative more energy-efficient heating systems	Slovakia	2021	Renewal Plan	Government of the Slovak Republic

Source: Authors' own elaboration.

Note: The original titles for each of the documents listed here are provided in the Annex Table 3. List of documents analysed.

There is also the more widely discussed theme of ii) consultations (3.4%) across institutions and stakeholders. Switzerland, a country known for widespread direct democracy processes, is also not surprisingly covering those topics at much larger scale than the average (5.2%). Yet, we see little variation on the minimum scale, pointing to the diffusion of joint governance beyond country specificities across all

countries. A case in point here is the consultation process with citizens Japan introduced on nuclear power risks (c. in Table 7).

Second, policy agility and setting directions in policy are other core themes in the total topics modelled, focusing on topics of monitoring and evaluation (2.6%) and foresight (2.5%) (Table 6). Specific examples discussed include building future scenarios for regional policies that factor in natural threats in Chile (e. in Table 7) and a focus on evaluation for Hungary's climate action plan (g. in Table 7).

Third, a topic that receives some coverage is directed policy towards green objectives. This includes, for instance, public procurement and regulatory measures for green housing in Korea (k. in Table 7) and France (j. in Table 7). Another area regards support for STI for more sustainable transportation in Costa Rica (i. in Table 7).

Furthermore, in accordance with the relatively high percentage of strategies allocated to traditional innovation policies by lower R&D spender countries (23.5%), this group accounts for the lowest percentage of new innovation policy approaches – 2%. Conversely, the top 10 R&D spenders hold a percentage share for traditional innovation of 3.4% and for new approaches of 11%.

5 Conclusions

This paper provides quantitative evidence of the prominence that Innovation Policy 3.0 goals have on governments' STI policy agenda across 24 OECD member countries. Green transition goals dominate the shifted STI policy agenda. Social and regional inclusion and digitalisation are also discussed as part of these priorities. Traditional innovation policy goals, which focus on firms, industry and growth and the research sector and its linkages with industry, are also dealt with in strategies but at less than half the prevalence. The strategies reviewed also incorporate elements related to new policy approaches required for STI policy to reach those goals, including joint governance, agility, and directionality.

From a methodological standpoint, this paper demonstrates the feasibility of conducting extensive assessments of qualitative policy data using the topic modelling approach, albeit with certain limitations. The role of conceptual thinking and expert analysis remains paramount, as the nuanced nature of language and policy discussions demand interpretation, setting it apart from the objectivity of quantitative data analysis. AI-based techniques, including the use of generative AI technologies, open even richer avenues going forward without negating, however, the importance of conceptual thinking and expert analysis.

References

- Acemoglu, D. and S. Johnson (2023), *Power and Progress: Our Thousand-Year Struggle Over Technology and Prosperity*, Public Affairs. [1]
- Arnold, E. et al. (2023), *Navigating Green and Digital Transitions: Five Imperatives for Effective STI Policy*, OECD Publishing, <https://doi.org/10.1787/dffb0747-en>. [18]
- Arts, S., J. Hou and J. Gomez (2021), “Natural language processing to identify the creation and impact of new technologies in patent text: Code, data, and new measures”, *Research Policy*, Vol. 50/2, p. 104144, <https://doi.org/10.1016/j.respol.2020.104144>. [13]
- Asheim, B., J. Moodysson and F. Tödtling (2011), “Constructing Regional Advantage: Towards State-of-the-Art Regional Innovation System Policies in Europe?”, *European Planning Studies*, Vol. 19/7, pp. 1133-1139, <https://doi.org/10.1080/09654313.2011.573127>. [5]
- Bas, M., A. Fernandes and C. Paunov (2023), “How resilient was trade to COVID-19?”, *Economics Letters*, 111080, Chicago. [28]
- Benoit, K. (2020), *Text as Data: An Overview*, https://kenbenoit.net/pdfs/CURINI_FRANZESE_Ch26.pdf. [12]
- Bischof, J. and E. Airoldi (2021), *Summarizing topical content with word frequency and exclusivity*. [27]
- Blei, D. (2012), “Probabilistic topic models”, *Communications of the ACM*, Vol. 55/4, pp. 77-84, <https://doi.org/10.1145/2133806.2133826>. [22]
- Boon, W. and J. Edler (2018), “Demand, challenges, and innovation. Making sense of new trends in innovation policy”, *Science and Public Policy*, Vol. 45/4, pp. 435-447, <https://doi.org/10.1093/scipol/scy014>. [8]
- Camarg, B. and A. Justo (2016), *IRaMuTeQ: R INTERFACE for multidimensional analysis of texts and questionnaires*, IRaMuTeQ, http://iramuteq.org/documentation/fichiers/IRaMuTeQ%20Tutorial%20translated%20to%20English_17.03.2016.pdf. [32]
- David Mimno, H. (2011), *Optimizing Semantic Coherence in Topic Models*, <https://aclanthology.org/D11-1024/>. [26]

- Diercks, G., H. Larsen and F. Steward (2019), "Transformative innovation policy: Addressing variety in an emerging policy paradigm", *Research Policy*, Vol. 48/4, pp. 880-894, <https://doi.org/10.1016/j.respol.2018.10.028>. [4]
- Gentzkow, M., B. Kelly and M. Taddy (2019), "Text as Data", *Journal of Economic Literature*, Vol. 57/3, pp. 535-574, <https://doi.org/10.1257/jel.20181020>. [11]
- Grillitsch, M. et al. (2019), "Innovation policy for system-wide transformation: The case of strategic innovation programmes (SIPs) in Sweden", *Research Policy*, Vol. 48/4, pp. 1048-1061, <https://doi.org/10.1016/j.respol.2018.10.004>. [3]
- Haddad, C. and A. Bergek (2023), "Towards an integrated framework for evaluating transformative innovation policy", *Research Policy*, Vol. 52/2, p. 104676, <https://doi.org/10.1016/j.respol.2022.104676>. [7]
- Hannigan, T. et al. (2019), "Topic Modeling in Management Research: Rendering New Theory from Textual Data", *Academy of Management Annals*, Vol. 13/2, pp. 586-632, <https://doi.org/10.5465/annals.2017.0099>. [23]
- Howlett, M., M. Ramesh and A. Perl (2020), *Studying public policy: Policy cycles and policy subsystems*, Toronto: Oxford University Press. [16]
- Laranja, M., E. Uyarra and K. Flanagan (2008), "Policies for science, technology and innovation: Translating rationales into regional policies in a multi-level setting", *Research Policy*, Vol. 37/5, pp. 823-835, <https://doi.org/10.1016/j.respol.2008.03.006>. [15]
- Lundvall, B. (2007), "National Innovation Systems—Analytical Concept and Development Tool", *Industry & Innovation*, Vol. 14/1, pp. 95-119, <https://doi.org/10.1080/13662710601130863>. [31]
- Mazzucato, M. (2016), "From market fixing to market-creating: a new framework for innovation policy", *Industry and Innovation*, Vol. 23/2, pp. 140-156, <https://doi.org/10.1080/13662716.2016.1146124>. [6]
- Nelson, R. (ed.) (1993), *National innovation systems: a comparative analysis*, Oxford University Press, USA, 1993. [29]
- OECD (2015), *Systems Innovation: Synthesis Report*, OECD Publishing, Paris. [17]
- OECD (1997), *National Innovation Systems*, OECD Publishing, Paris. [30]
- Paunov, C. and S. Planes-Satorra (2021), "Science, technology and innovation in the time of COVID-19", *OECD Science, Technology and Industry Policy Papers*, No. 99, OECD Publishing, Paris, <https://doi.org/10.1787/234a00e5-en>. [2]
- Roberts, M., B. Stewart and D. Tingley (2019), "stm: An R Package for Structural Topic Models", *Journal of Statistical Software*, Vol. 91/2, <https://doi.org/10.18637/jss.v091.i02>. [24]
- Roberts, M. et al. (2014), "Structural Topic Models for Open-Ended Survey Responses", *American Journal of Political Science*, Vol. 58/4, pp. 1064-1082, <https://doi.org/10.1111/ajps.12103>. [25]

- Rüegg-Stürm, J. (2005), *The New St. Gallen Management Model*, Palgrave Macmillan UK, London, <https://doi.org/10.1057/9780230505162>. [20]
- Schlaile, M. et al. (2017), "Innovation Systems for Transformations towards Sustainability? Taking the Normative Dimension Seriously", *Sustainability*, Vol. 9/12, p. 2253, <https://doi.org/10.3390/su9122253>. [9]
- Schot, J. and W. Steinmueller (2016), *Framing innovation policy for transformative change: Innovation policy 3.0*, SPRU Science Policy Research Unit, University of Sussex: Brighton, UK. [14]
- Wauters, B. (2019), *Strategic management in the public sector and public policy-making: friend or foe?*, European Social Fund Public Administration and Governance Network, https://www.researchgate.net/publication/333774591_Strategic_management_in_the_public_sector_and_public_policy-making_friend_or_foe (accessed on 16 November 2021). [19]
- Weber, K. and H. Rohrer (2012), "Legitimizing research, technology and innovation policies for transformative change", *Research Policy*, Vol. 41/6, pp. 1037-1047, <https://doi.org/10.1016/j.respol.2011.10.015>. [10]
- Weiser, A. et al. (2020), "Towards a more sustainable metal use – Lessons learned from national strategy documents", *Resources Policy*, Vol. 68, p. 101770, <https://doi.org/10.1016/j.resourpol.2020.101770>. [21]

Annex A. Pre-processing and cleaning steps for the topic modelling

To prepare our database for the text analysis, we applied several **pre-processing steps**. The documents were trimmed to remove page margins containing page numbers, repeated references to titles, chapters, or the like. The files were then converted from PDF to TXT format, which removes all graphics and formatting from the document, leaving only plain text. Where documents were not provided in English, they were translated using Google Translate.

Taking the TXT file for each document, we conducted four basic text **cleaning steps**:

Lemmatise words. We reduce words to their “lemma” – their base form without any conjugation or pluralisation and expand acronyms like “AI” (artificial intelligence) to their full written form.

Remove stop words. We remove prepositions, articles, and conjunctions that serve a grammatical function but do not contain any semantic information. Common stop words include “the”, “and”, and “of”. Numbers, misspelled words, and non-English words are all removed from the corpus.

- **Remove punctuation.** Punctuation marks are removed at the end of the process. Line breaks are retained, as these prevent concepts from different paragraphs from being mixed incorrectly.
- **Identify compounds.** The lemmatised text is analysed for compound expressions. “Climate change”, for example, is a compound word composed of “climate” and “change”, which takes on a different meaning than the two words individually.

Annex Table 1. Topics, themes and categories in the corpus

	Number / Topic label	Common words	FREX words
Category: Context			
Theme: Crisis recovery and resilience			
1	COVID-19 and resilience	impact, covid, economic, social, pandemic, crisis, resilience	covid, pandemic, crisis, shock, health_crisis, territorial_cohesion, economic_recovery
Theme: Energy prices			
2	Global energy markets	market, demand, large, supply, price, small, country	market, demand, price, import, service_sector, crude_oil_price, barrel
Category: Traditional innovation policy			
Theme: Firms, industry and growth			
3	Businesses	company, make, business, made, form, means, hand	company, make, SMEs, player, choice, good, large_company
4	Growth	increase, increased, gdp, expected, average, result, number	gdp, average, percent, percentage, index, percentage_point, increased
5	Conditions for growth	support, aim, order, addition, region, country, strengthening	aim, strengthening, promoting, promotion, improving, long_term, human_resource
6	Investment and productivity	investment, sector, economy, term, growth, significant, high	private_investment, public_investment, boosting, productivity, job_creation, labour_productivity, economic_growth

Theme: Supporting industry-science linkages			
7	Research	research, innovation, r_d, university, science, researcher, collaboration	research, researcher, excellence, research_infrastructure, research_result, research_centre, applied_research
8	Support for industry-science linkages	support, work, community, industry, programme, opportunity, business	engagement, commercialisation, emerging_technology, responsive, interagency, suite, research_enterprise
Category			
Theme: Green transition goals			
9	Climate change and adaptation policy	measure, plan, change, climate, climate_change, goal, adopted	climate, adaptation, climate_policy, climate_protection, adaptation_measure, climate_change, effect_of_climate
10	Pollution mitigation	project, operation, end, start, management, construction, supported	project, operation, site, radioactive_waste, spent_fuel, call, basin
11	Power generation	base, system, addition, effort, business, introduction, power_generation	power_source, business_operator, stable_supply, solar_power_generation, thermal_power_generation, distribution_business_operator, adjustment_power
12	Energy efficiency	technology, equipment, utilisation, development, field, industry, demonstration	demonstration, energy_saving, smart_city, social_implementation, next_generation, convenience, fuel_efficiency
13	Emissions	energy, emission, hydrogen, co2, renewable, electricity, fuel	co2, renewable, heat, energy_source, carbon, fossil_fuel, biomass
14	Clean mobility	building, transport, vehicle, mobility, energy, construction, renovation	renovation, road, rail, public_transport, fleet, buses, passenger
15	Sustainable industries	sustainable, innovation, industry, economic, transition, society, country	bioeconomy, heritage, tourist, cultural_heritage, environmental_sustainability, social_innovation, creative_industry
Theme: Social and regional inclusion			
20	Inclusion in the labour market	employment, people, women, labour_market, work, child, worker	women, worker, family, man, disability, gender, poverty
21	Capacity-building	training, education, skill, student, school, programme, learning	training, education, skill, student, school, teacher, educational
22	Regional development	area, resource, related, condition, regional, activities, local	local, municipality, regional, area, resource, condition, spatial_planning
Theme: Digital transformation goals			
16	Technologies for industry 4.0 and defence	technology, security, applications, space, artificial_intelligence, software, capability	artificial_intelligence, quantum, defence, intelligence, sensor, robotics, attack
17	Digital infrastructure	service, digital, infrastructure, citizen, network, access, component	digitization, digital_transformation, public_service, connectivity, digital_infrastructure, digital_service, broadband
18	Data	data, information, platform, user, applications, online, public	database, medical, hospital, interface, portal, blockchain, web
Theme: Health knowledge and technology			
19	Health knowledge and technology	world, people, society, technology, human, country, knowledge	disease, therapy, genetic, brain, understand, tissue, human
Category			
Theme: Engagement			
23	Vertical governance	government, ensure, support, continue, funding, programme, federal	government, province, federal_government, federal, provincial, ongoing, department
24	Consultation	organisation, work, group, practise, institution, involved, agreement	involved, consultation, expert, representative, working_group, involvement, association
Theme: Agility			
25	Monitoring and evaluation	implementation, evaluation, monitoring, programme, result, carried, document	implementation, execution, preparation, coordination, monitoring, operational_programme, evaluation
26	Foresight	account, target, period, base, reduction, scenario, costs	target, scenario, taking, account, estimate, indicative, projection
Theme: Directional policy			
27	Directional policy	future, focus, important, challenge, goal, potential, approach	broad, focus, funding_programme, societal, shaping, important_role, field_of_action

Annex Table 2. Topics, themes identified in the misc. category

Number / Topic label		Common words	FREX words
Theme: Strategy and procedural aspects			
28	Design	development, system, process, base, field, tool, design	design, tool, integrated, process, system, development, technical
29	Strategy	national, strategy, policy, objective, priority, set, framework	rdi, national, sectoral, actor, action_plan, theme, instrument
30	Documents	report, section, review, authority, information, proposal, performance	report, issued, disclosure, submitted, officer, subsection, annual report
31	Location and time context	time, place, state, structure, risk, existing, general	place, structure, possibility, state, created, view, time
Theme: Government action			
32	Policy intervention	action, objective, aimed, territory, context, intervention, guarantee	transversal, articulation, public_policy, line_of_action, incorporation, favour, planning_instrument
33	Tax	tax, amount, payment, revenue, grant, employee, benefit	tax, payment, loan, expense, insurance, liability, rent
34	Regulatory and legal reform	reform, procedure, control, law, ensure, legislation, investment	court, corruption, simplification, judge, money_laundering, state_aid, judiciary
35	Regulation	regulation, order, requirement, standard, principle, established, framework	regulation, regulatory, principle, requirement, legal, standard, criteria

Annex Table 3. List of documents analysed

Country	Document type	Nb.	Reference (English title [Original title], (Institution, Year))
Australia	Ministerial plan	1.	COVID-19: Recovery and resilience (Australia's National Science Agency, 2020)
		2.	CSIRO Corporate Plan 2019-2020 (Australia's National Science Agency, 2020)
		3.	Partnerships for Recovery: Australia's COVID-19 Development Response (Department of Foreign Affairs and Trade, 2020)
		4.	CSIRO Corporate Plan 2020-2021 (Australia's National Science Agency, 2021)
	Government strategy	5.	Australian Medical Research and Innovation Strategy 2016-2021 (Australian Government, 2016)
		6.	Data Strategy 2018-2020 (Department of Industry, Innovation and Science, 2018)
		7.	Digital Transformation Agency: Vision 2025 (Digital Transformation Agency, 2018)
		8.	Australia's National Hydrogen Strategy (COAG Energy Council, 2019)
		9.	Budget 2020-2021 (Australian Government, 2020)
		10.	Digital Economy Strategy 2030 (Australian Government, 2021)
		11.	Make It Happen: The Australian Government's Modern Manufacturing Strategy (Australian Government, 2021)
	STI strategy	12.	National Innovation and Science Agenda (Australian Government, 2015)
		13.	Global Innovation Strategy (Australian Government, 2016)
		14.	Technology Investment Roadmap (Department of Industry, Science, Energy and Resources, 2020)
Austria	Government strategy	15.	Open Innovation Strategy for Austria [Open Innovation Strategie für Österreich] (Federal Ministry of Science and Research, 2016)
		16.	Bioeconomy: A Strategy for Austria (Ministry for Sustainability and Tourism, 2019)
		17.	National Reform Programme 2019 (Federal Chancellery, 2019)
		18.	Digital Action Plan Austria [Digitaler Aktionsplan Austria] (Federal Chancellery, 2020)
		19.	STI Strategy for Mobility [FTI-Strategie Mobilität] (Federal Ministry for Climate Action, Environment, Energy, Mobility, Innovation and Technology, 2020)
		20.	National Reform Programme 2020 (Federal Chancellery, 2020)
		21.	Recovery and Resilience Plan [Österreichischer Aufbau- und Resilienzplan 2020-2026] (Federal Chancellery, 2021)

		22.	Strategy of the Federal Government for AI [Strategie der Bundesregierung für Künstliche Intelligenz] (Federal Chancellery, 2021)
		23.	Circular Economy Strategy [Kreislaufwirtschafts-Strategie] (Federal Chancellery, 2021)
	STI strategy	24.	STI Strategy 2030 [FTI-Strategie 2030] (Federal Government, 2020)
Belgium	Government strategy	25.	Belgian National Adaptation Plan (Federal Government, 2017)
		26.	National Energy-Climate Plan [Plan national énergie-climat 2021-2030] (National Climate Commission, 2019)
		27.	National Reform Programme 2019 (Federal Government, 2019)
		28.	Government Agreement [Accord de gouvernement] (Federal Government of Belgium, 2020)
		29.	National Reform Programme 2020 (Federal Government of Belgium, 2020)
		30.	Cabinet of the State Secretary for Restart and Strategic Investment [Plan national pour la reprise et la resilience] (National Plan for Restart and Resilience, 2021)
		31.	Scenarios for a climate neutral Belgium by 2050 (DG Environment, 2021)
		32.	ISED Departmental Plan 2019-20 (Innovation, Science and Economic Development Canada, 2019)
		33.	Responding to Canada's Needs from COVID-19 to Climate Change (National Research Council Canada, 2020)
Canada	Ministerial plan	34.	NRC Strategic Plan 2019-2024 (National Research Council Canada, 2020)
		35.	NSERC Departmental Plan 2020-21 (Natural Sciences and Engineering Research Council of Canada, 2020)
		36.	CIHR Departmental Plan 2020-21 (Canadian Institutes of Health Research, 2020)
		37.	ISED Departmental Plan 2021-22 (Innovation, Science and Economic Development Canada, 2021)
		38.	NSERC Departmental Plan 2021-22 (Natural Sciences and Engineering Research Council of Canada, 2021)
		39.	CIHR Strategic Plan 2021-2031 (Canadian Institutes of Health Research, 2021)
		40.	CIHR Departmental Plan 2021-22 (Canadian Institutes of Health Research, 2021)
	Government strategy	41.	Digital Operations Strategic Plan: 2018-2022 (Treasury Board of Canada Secretariat, 2018)
		42.	Building a Nation of Innovators (Innovation, Science and Economic Development Canada, 2019)
		43.	Investing in the Middle Class: Budget 2019 (Government of Canada, 2019)
		44.	Supporting Canadians and Fighting COVID-19: Fall Economic Statement (Government of Canada, 2020)
		45.	A Healthy Environment and a Healthy Economy (Environment and Climate Change Canada, 2020)
		46.	Canada's Climate Actions (Environment and Climate Change Canada, 2021)
		47.	Digital Operations Strategic Plan: 2021-2024 (Treasury Board of Canada Secretariat, 2021)
		48.	A Recovery Plan for Jobs, Growth, and Resilience: Budget 2021 (Government of Canada, 2021)
		49.	Intellectual Property Strategy (Innovation, Science and Economic Development Canada, 2021)
	STI strategy	50.	Seizing Canada's Moment: Moving Forward in Science, Technology and Innovation (Industry Canada, 2014)
		51.	Canada's Science Vision (Innovation, Science and Economic Development Canada, 2019)
		52.	Restart, Recover and Reimagine Prosperity for all Canadians (Industry Strategy Council, 2020)
Chile	Government strategy	53.	Report Towards a Disaster Resilient Chile: An Opportunity [Informe Hacia un Chile Resiliente frente a Desastres: Una Oportunidad] (National Council for Innovation for Development, 2016)
		54.	Science and Innovation for Chile's Water Challenges Report [Informe Ciencia e Innovación para los Desafíos del Agua en Chile] (National Council for Innovation for Development, 2017)
		55.	Energy 2050: Chile's Energy Policy [Energía 2050: Política Energético de Chile] (Ministerio de Energía, 2017)
		56.	National Artificial Intelligence Policy [Política Nacional de Inteligencia Artificial] (Ministry of Science, Technology and Innovation, 2019)
		57.	Proposal for a Policy on Open Access to scientific information and publicly-funded research data from ANID [Propuesta de Política de acceso abierto a la información científica y adatos de investigación financiados con fondos públicos de la ANID] (Ministry of Science, Technology and

	STI strategy		Innovation, 2020)
		58.	A common dream for the future of Chile: Report for the President of the Republic, Michelle Bachelet [Un sueño compartido para el futuro de Chile: Informe a la Presidenta de la República, Michelle Bachelet] (Presidential Commission Science for Development of Chile, 2015)
		59.	Science, technology and innovation in a new pact on sustainable and inclusive development: strategic orientations in view of 2030 after a 10-years trajectory [Ciencias, tecnologías e innovación para un nuevo pacto de desarrollo sostenible e inclusivo: Orientaciones estratégicas de cara a 2030 tras diez años de trayectoria] (National Council for Innovation for Development, 2017)
		60.	Science, Technology, Knowledge and Innovation in Chile [Ciencia, Tecnología, Conocimiento e Innovación para Chile] (National Council for Innovation for Development, 2019)
		61.	National Policy on Science, Technology, Knowledge and Innovation [Política nacional de ciencia, tecnología, conocimiento e innovación] (Ministry of Science, Technology and Innovation, 2020)
		62.	Action Plan: National Policy on Science, Technology, Knowledge and Innovation [Plan de acción: Política nacional de ciencia, tecnología, conocimiento e innovación] (Ministry of Science, Technology and Innovation, 2021)
		63.	Basis for National Science, Technology, Knowledge and Innovation Strategy [Base para la Estrategia Nacional de Ciencia, Tecnología, Conocimiento e Innovación] (National Council on Innovation for Development, 2021)
		64.	National Gender Equality Policy in CTCI [Política Nacional de Igualdad de Género en CTCI] (Ministry of Science, Technology and Innovation, 2021)
		65.	Technology Development and Transfer Strategy for Climate Change [Estrategia de Desarrollo y Transferencia Tecnológica para el Cambio Climático] (Ministry of Science, Technology and Innovation, 2021)
Costa Rica	Legislation	66.	Law 9971 Creation of the Costa Rican Research and Innovation Promotion Agency. [Ley 9971 Creación de la Promotora Costarricense de Investigación e Innovación] (Ministry of Science, Technology and Telecommunication, 2021)
		67.	Law 9960 Creation of the Costa Rican Space Agency [Ley 9960 Creación de la Agencia Espacial Costarricense] (Ministry of Science, Technology and Telecommunication, 2021)
	Ministerial plan	68.	National Entrepreneurship Policy 2018-2022 [Política Nacional de Empresariedad 2018-2022] (Ministry of Economics, Industry and Commerce, 2018)
	Government strategy	69.	National Telecommunications Development Plan (Ministry of Science, Technology and Telecommunication, 2015)
		70.	National Decarbonisation Plan 2018-2050 [Plan Nacional de Descarbonización 2018-2050] (Ministry of Environment and Energy, 2018)
		71.	Digital Transformation Strategy towards Costa Rica of the Bicentennial 4.0 [Estrategia de Transformación Digital hacia la Costa Rica del Bicentenario 4.0] (National Government; Ministry of Science, Technology and Telecommunication, 2018)
		72.	Territorial Economic Strategy for an Inclusive and Decarbonised Economy 2020-2050 [Estrategia Económica Territorial para una Economía Inclusiva y Descarbonizada 2020-2050] (Ministry of National Planification and Economic Policy, 2020)
		73.	National Knowledge-Based Economy and Society Policy 2022-2050 [Política Nacional De Sociedad y Economía Basadas en el Conocimiento 2022-2050] (Ministry of Science, Technology and Telecommunication, 2021)
		74.	Methodological Guide for the elaboration and design of the National Telecommunications Development Plan 2022-2027 [Guía Metodológica para la elaboración y diseño del Plan Nacional de Desarrollo de Telecomunicaciones 2022-2027] (Ministry of Science, Technology and Telecommunication, 2021)
		75.	Strategy for the Responsible and Ethical Use of AI [Estrategia para el uso responsable y ético de la IA] (Ministry of National Planification and Economic Policy, 2021)
	STI strategy	76.	Route 2021: Knowledge and Innovation for Competitiveness, Prosperity, and Well-being [Ruta 2021: Conocimiento e Innovación para la Competitividad, Prosperidad y Bienestar] (Ministry of Science, Technology, and Telecommunication, 2014)
		77.	National Plan for Science, Technology, and Innovation 2015 - 2021 [Plan Nacional de Ciencia, Tecnología e Innovación 2015 - 2021] (Ministry of Science, Technology, and Telecommunication, 2015)
		78.	Institutional Strategic Plan 2019-2025 [Plan estratégico institucional 2019-2025] (Ministry of

			Science, Technology and Telecommunication; Secretary of Institutional and Sectorial Planification, 2019)
		79.	National Policy on Knowledge-Based Economy and Society and National Plan for Science, Technology and Innovation [Política Nacional de Economía y Sociedad Basada en el Conocimiento y Plan Nacional de Ciencia Tecnología e Innovación] (Ministry of Science, Technology and Telecommunication, 2021)
		80.	National Science, Technology and Innovation Plan 2022-2027 [Plan Nacional de Ciencia, Tecnología e Innovación 2022-2027] (Ministry of Science, Technology and Telecommunication, 2021)
		81.	National Bio-economy Strategy 2020-2030 [Estrategia Nacional de Bioeconomía 2020-2030] (Ministry of Science, Technology and Telecommunication, 2020)
Finland	Government strategy	82.	Inclusive and Competent Finland [Inclusive and Competent Finland] (Finnish Government, 2019)
		83.	Finland's National Reform Programme 2019 (Ministry of Finance, 2019)
		84.	The regional development decision 2020–2023: Sustainable and vital regions (Ministry of Economic Affairs and Employment, 2020)
		85.	Government Report on the Implementation of the 2030 Agenda (Prime Minister's Office, 2020)
		86.	National Reform Programme 2020 (Ministry of Finance, 2020)
		87.	Finland's sustainable growth agenda: preliminary recovery and reconstruction plan [Suomen kestävä kasvun ohjelma – alustava elpymisja palautumissuunnitelma] (Ministry of Finance, 2021)
France	STI strategy	88.	Vision and road map of the Research and Innovation Council Finland (Research and Innovation Council, 2017)
		89.	RDI Roadmap: Solutions for a sustainable and developing society (State Council, 2020)
	Ministerial plan	90.	Strategic Ambitions 2020-2023 [Ambitions Stratégiques 2020-2023] (BpiFrance, 2020)
		91.	2019: A Year with the CNRS [2019: Une Année avec le CNRS] (CNRS, 2020)
		92.	2020: A Year with the CNRS [2020: Une Année avec le CNRS] (CNRS, 2021)
	Government strategy	93.	National Reform Programme 2019 (Prime Minister of France, 2019)
		94.	Support Plan for Technology Companies [Plan de Soutien aux Entreprises Technologiques] (Ministry of Economy and Finance, 2020)
		95.	National Reform Programme 2020 [Programme national de réforme 2020] (Prime Minister of France, 2020)
		96.	National Recovery and Resilience Plan [Plan National de Relance et de Résilience] (Government of France, 2021)
		97.	What Place for Skills in Business? New Approaches for Changing Practices [Quelle place pour les compétences dans l'entreprise? Renouveler les approches pour refonder les pratiques] (Employee Skill Network, 2021)
	STI strategy	98.	A Strategic Agenda for Research, Technology Transfer, and Innovation (Ministry of Higher Education and Research, 2013)
		99.	Innovation: A Major Challenge for France [L'Innovation, un enjeu majeur pour la France] (Ministry for a Productive Recovery & Ministry of Higher Education and Research, 2013)
		100.	Fifteen years of innovation policy in France [Quinze ans de politiques d'innovation en France] (National Commission for Innovation Policy Evaluation, 2016)
Germany	Government strategy	101.	Framework programme for sustainable development (FONA) [Rahmenprogramm Forschung für Nachhaltige Entwicklung (FONA)] (Federal Ministry for Education and Research, 2015)
		102.	Health Research Framework Programme [Rahmenprogramm Gesundheitsforschung] (Federal Government, 2018)
		103.	Artificial Intelligence Strategy [Strategie Künstliche Intelligenz] (Federal Government, 2018)
		104.	Quantum technologies - from the basics to the market [Quantentechnologien - von den Grundlagen zum Markt] (Federal Ministry for Economic Affairs and Climate Action, 2018)
		105.	STEM Action Plan [MINT-Aktionsplan] (Federal Ministry for Education and Research, 2019)
		106.	National Further Education Strategy [Nationale Weiterbildungsstrategie] (Federal Ministry for Education and Research, 2019)
		107.	Action plan "Research for autonomous driving - an overarching research framework by BMBF, BMWi and BMVI". [Aktionsplan „Forschung für autonomes Fahren – ein übergreifender Forschungsrahmen von BMBF, BMWi und BMVI“] (Federal Ministry for Economic Affairs and

			Climate Action, 2019)
		108.	Research Factory Battery [Dachkonzepts "Forschungs-fabrik Batterie"] (Federal Ministry for Economic Affairs and Climate Action, 2019)
		109.	Blockchain Strategy [Blockchain-Strategie] (Federal Ministry for Economic Affairs and Climate Action, 2019)
		110.	Digital Strategy of the BMBF "Digital Future: Learning. Research. Knowledge. [Digitalstrategie des BMBF „Digitale Zukunft: Lernen. Forschen. Wissen.“] (Federal Ministry for Education and Research, 2019)
		111.	Tackling consequences of the pandemic, securing prosperity, strengthening future prospects [Corona-Folgen bekämpfen, Wohlstand sichern, Zukunftsfähigkeit stärken] (Federal Coalition Committee, 2020)
		112.	From the Idea to Market Success: Programmes for innovative SMEs [Von der Idee zum Markterfolg: Programme für einen innovativen Mittelstand] (Federal Ministry for Economic Affairs and Climate Action, 2020)
		113.	Research for Sustainability Strategy (Federal Government, 2020)
		114.	National Bioeconomy Strategy [Nationale Bioökonomiestrategie] (Federal Government, 2020)
		115.	The National Reform Programme 2020 [Das Nationale Reformprogramm 2020] (Federal Government, 2020)
		116.	Artificial Intelligence Strategy - Update [Strategie Künstliche Intelligenz – Fortschreibung] (Federal Government, 2020)
		117.	Nationale Wasserstoffstrategie [Nationale Wasserstoffstrategie] (Federal Government, 2020)
		118.	German Recovery and Resilience Plan (Federal Ministry of Finance, 2021)
		119.	National Further Education Strategy [Nationale Weiterbildungsstrategie] (Federal Ministry for Education and Research, 2021)
		120.	National Reform Programme 2021 [Das Nationale Reformprogramm 2021] (Federal Ministry for Economic Affairs and Climate Action, 2021)
		121.	Roadmap Quantencomputing [Roadmap Quantencomputing] (Federal Ministry for Economic Affairs and Climate Action, 2021)
		122.	Shaping the future with technological sovereignty. BMBF Impulse Paper on Technological Sovereignty [Technologisch souverän die Zukunft gestalten. BMBF-Impulspapier zur technologischen Souveränität] (Federal Ministry for Education and Research, 2021)
	STI strategy	123.	Internationalisation of education, science and research [Internationalisierung von Bildung, Wissenschaft und Forschung] (Federal Government, 2017)
		124.	High-Tech Strategy 2025 [Hightech-Strategie 2025] (Federal Government, 2018)
Greece	Legislation	125.	Detailed Description of Actions [Αναλυτική Περιγραφή Δράσεων] (Government of Greece, 2021)
	Ministerial plan	126.	Working in the Public Sector through international practice in the era of COVID-19 [Η εργασία στον Δημόσιο Τομέα μέσα από την διεθνή πρακτική την εποχή του COVID-19] (Ministry of Interior – Innovation Unit, 2020)
	Government strategy	127.	Corporate Pact for the Development Framework - NSRF 2014-2020 [Εταιρικό Σύμφωνο για το Πλαίσιο Ανάπτυξης – ΕΣΠΑ 2014-2020] (General Secretariat for Public Investment, 2014)
		128.	National Research and Innovation Strategy for Smart Specialisation 2014-2020 [Εθνική Στρατηγική Έρευνας και Καινοτομίας για την Έξυπνη Εξειδίκευση 2014-2020] (Ministry of Economy, Infrastructure, Shipping and Tourism, 2014)
		129.	Greece: National Strategy for Sustainable and and Fair Development 2030 [Ελλάδα: Εθνική Στρατηγική για τη Βιώσιμη και Δίκαιη Ανάπτυξη 2030] (Ministry of Economy and Development, 2019)
		130.	National Reform Programme 2019 (Government of Greece, 2019)
		131.	Development Plan for the Greek Economy [Σχέδιο Ανάπτυξης για την Ελληνική Οικονομία] (European Commission, 2020)
		132.	National Recovery and Resilience Plan [Εθνικό Σχέδιο Ανακαμψης Και Ανθεκτικότητας] (Ministry of Finance, 2021)
		133.	Partnership Agreement - Republic of Greece [Συμφωνία εταιρικής σχέσης - Ελληνική Δημοκρατία] (Government of the Hellenic Republic, 2021)
Hungary	Government strategy	134.	A Change of Pace in Higher Education (Ministry of Human Capacities, 2016)
		135.	OGY Decision on the second National Climate Change Strategy 2018-2030 looking ahead to 2050 [OGY határozat a 2018-2030 közötti időszakra vonatkozó, 2050-ig tartó időszakra

			kitekintést nyújtó második Nemzeti Éghajlatváltozási Stratégiáról] (Ministry of Innovation and Technology, 2018)
		136.	National Reform Programme 2019 [2019. évi Nemzeti Reform Programja] (Government of Hungary, 2019)
		137.	The Strategy for Strengthening Hungarian Micro, Small and Medium Enterprises [Magyarország középtávú? Kutatás stratégiája.] (Ministry of Innovation and Technology, 2019)
		138.	The Strategy for Strengthening Hungarian Micro, Small and Medium Enterprises [A Magyar Mikro-, Kis- És Középvállalkozások Megerősítésének Stratégiája] (Ministry of Innovation and Technology, 2019)
		139.	Climate Change Action Plan [Éghajlatváltozási Cselekvési Terv] (Ministry of Innovation and Technology, 2020)
		140.	2020 National Reform Programme of Hungary (Government of Hungary, 2020)
		141.	National Digitisation Strategy 2021-2030 [Nemzeti Digitalizációs Stratégia 2021-2030] (National Research, Development and Innovation Office, 2020)
		142.	Hungary's Artificial Intelligence Strategy 2020-2030 (Ministry for Innovation and Technology, 2020)
		143.	Hungary's Recovery and Control Plan [Magyarország Helyreállítási és Ellenállókepeségi Terve] (Government of Hungary, 2021)
		144.	Hungary's National Reform Programme for 2021 [Magyarország 2021. évi Nemzeti Reform Programja] (Government of Hungary, 2021)
	STI strategy	145.	National Smart Specialisation Strategy (Ministry of Innovation and Technology, 2014)
		146.	National Research Infrastructure Roadmap (Ministry of Innovation and Technology, 2018)
		147.	Investing in Knowledge, Investing in the Future [Befektetés A Tudásba, Befektetés A Jövőbe] (National Research, Development and Innovation Office, 2018)
		148.	National Strategy for Smart Specialisation [Nemzeti Intelligens Szakosodási Stratégia] (Ministry of Innovation and Technology, 2021)
		149.	Hungary's Research, Development and Innovation Strategy [Magyarország Kutatási, Fejlesztési és Innovációs Stratégiája] (Ministry of Innovation and Technology, 2021)
Ireland	Legislation	150.	General Scheme of the Higher Education Authority (Government of Ireland, 2021)
	Ministerial plan	151.	Science Foundation Ireland Annual Plan 2020 (Science Foundation Ireland, 2020)
		152.	Enterprise Ireland's Strategic Priorities 2021 (Enterprise Ireland, 2021)
		153.	Statement of Strategy 2021–2023 (Science Foundation Ireland, 2021)
	Government strategy	154.	National Reform Programme April 2019 (Government of Ireland, 2019)
		155.	National Reform Programme for the European Semester April 2020 (Government of Ireland, 2020)
		156.	Irish Research Council Strategic Plan 2020-2024 (Irish Research Council, 2020)
		157.	National Recovery and Resilience Plan 2021 (Government of Ireland, 2021)
		158.	Economic Recovery Plan (Government of Ireland, 2021)
		159.	AI – Here for Good: National AI Strategy for Ireland (Department of Enterprise, Trade and Employment, 2021)
	STI strategy	160.	Innovation 2020 - Excellence, Talent, Impact: Ireland's Strategy for Research and Development, Science, and Technology (Science Foundation Ireland, 2015)
		161.	Investing in Business, Enterprise and Innovation 2018-2027 (Government of Ireland, 2018)
		162.	Research Priority Areas 2018-2023 (Government of Ireland, 2018)
		163.	National Research and Innovation Strategy 2021-27 Consultation Paper (Government of Ireland, 2021)
		164.	Shaping Our Future Science Foundation Ireland Strategy 2025 (Science Foundation Ireland, 2021)
Italy	Ministerial plan	165.	Piano Nazionale Industria 4.0 [Piano Nazionale Industria 4.0] (Ministry of economic development, 2016)
		166.	National Strategy for Sustainable Development [Strategia Nazionale per lo sviluppo sostenibile] (Ministry of education, university and research; Ministry of economic development, 2017)
	Government	167.	Proposals for an Italian Artificial Intelligence Strategy [Proposte per una Strategia italiana per

	strategy		l'intelligenza artificiale] (Ministry of economic development, 2020)
		168.	The Three-Year Plan for Information Technology in Public Administration [Il Piano Triennale per l'informatica nella Pubblica Amministrazione] (Agency for Digital Italy; Department for Digital Transformation, 2020)
		169.	National Recovery and Resilience Plan [Piano Nazionale di Ripresa e Resilienza] (Council of Ministers, 2021)
	STI strategy	170.	Horizon 2020 (Ministry of Education, University and Research, 2013)
		171.	National Research Programme 2015-2020 [Programma Nazionale per la Ricerca 2015-2020] (Ministry of university and research, 2016)
		172.	National Strategy for Smart Specialisation [Strategia nazionale di specializzazione intelligente] (Ministry of the environment and protection of territory and sea, 2016)
		173.	National Programme for Research Infrastructures 2014-2020 [Programma Nazionale per le Infrastrutture di Ricerca 2014-2020] (Ministry of university and research, 2016)
		174.	Strategy for technological innovation and digitisation of the country 2025 [Strategia per l'innovazione tecnologica e la digitalizzazione del Paese 2025] (Ministry of technological innovation and digital transition, 2019)
		175.	National Research Programme 2021-2027 [Programma Nazionale per la Ricerca 2021-2027] (Ministry of university and research, 2020)
Japan	Legislation	176.	Basic Law on Science, Technology and Innovation [科学技術・イノベーション基本法] (Parliament of Japan, 1995)
		177.	Act on Stimulating the Creation of Science, Technology, and Innovation (Partial Amendment to the Act on Strengthening Research and Development Capability) [科学技術・イノベーション創出の活性化に関する法律（研究開発力強化法の一部改正）] (Parliament of Japan, 2018)
		178.	Basic Act on Science, Technology and Innovation (Amendment of the Basic Act on Science, Technology and Innovation). [科学技術・イノベーション基本法（科学技術基本法の改正）] (Parliament of Japan, 2020)
		179.	Partial amendment to the Act on the Stimulation of Science, Technology and Innovation Creation. [科学技術・イノベーション創出の活性化に関する法律の一部改正] (Parliament of Japan, 2020)
	Government strategy	180.	Strategic Energy Plan (2018)
		181.	Changes to the Declaration on the Creation of the World's Most Advanced Digital Nation and the Basic Plan for the Promotion of the Use of Public and Private Data [世界最先端デジタル国家創造宣言・官民データ活用推進基本計画の変更について] (IT Comprehensive Strategy Office, 2019)
		182.	AI Strategy 2019 (Cabinet Office, 2019)
		183.	Emergency Economic Measures to Cope with the Novel Coronavirus (COVID-19) (Cabinet Office, 2020)
		184.	Comprehensive Economic Measures to Secure People's Lives and Livelihoods toward Relief and Hope (Cabinet Office, 2020)
		185.	Annual report on energy [エネルギーに関する年次報告] (Ministry of Economy, Trade and Industry (METI), 2020)
		186.	Basic policy on reforms for the digital society [デジタル社会の実現に向けた改革の基本方針] (Cabinet Office, 2020)
		187.	Economic Measures for Overcoming COVID-19 and Opening Up a New Era (Cabinet Office, 2021)
		188.	Energy situation and main measures. [エネルギーをめぐる状況と主な対策] (Ministry of Economy, Trade and Industry (METI), 2021)
		189.	Green growth strategy associated with carbon neutrality in 2050. [2050年カーボンニュートラルに伴うグリーン成長戦略] (Ministry of Economy, Trade and Industry (METI), 2021)
		190.	Basic Energy Plan [エネルギー基本計画] (Ministry of Economy, Trade and Industry (METI), 2021)
		191.	Basic policy for Green Innovation Fund projects [グリーンイノベーション基金事業の基本方針] (Prime Minister's Office, 2021)
		192.	Draft global warming action plan [地球温暖化対策計画（案）] (Ministry of Economy, Trade and Industry (METI), 2021)

		193.	National Data Strategy (Digital Agency, 2021)
		194.	Semiconductor and digital industry strategy [半導体・デジタル産業戦略] (Ministry of Economy, Trade and Industry (METI), 2021)
	STI strategy	195.	Science and Technology Basic Plan [科学技術基本計画] (Cabinet Office, 2016)
		196.	Integrated Innovation Strategy 2019 [統合イノベーション戦略 2019] (Cabinet Office, 2019)
		197.	Integrated Innovation Strategy 2020 (Cabinet Office, 2020)
		198.	Industrial Technology Vision 2020. [産業技術ビジョン2020] (Cabinet Office, 2020)
		199.	Integrated innovation strategy 2021 [統合イノベーション戦略 2021] (Cabinet Office, 2021)
		200.	6th Science, Technology, and Innovation Basic Plan (Cabinet Office, 2021)
Korea	Ministerial plan	201.	2021 Work Plan [과학기술정보통신부2021년도 업무계획] (Ministry of Science and ICT, 2021)
		202.	Comprehensive Implementation Plan 2021 [2021년도 과학기술정보통신부연구개발사업 종합시행계획] (Ministry of Science and ICT, 2021)
	Government strategy	203.	National Strategy for Artificial Intelligence (Government of Korea, 2019)
		204.	Korea's Great Leap Forward as a Leader in Comprehensive Planning for the "Korean New Deal" [「한국판 뉴딜」 종합계획선도국가로 도약하는 대한민국 대전환] (Ministry of Planning and Finance, 2020)
		205.	2050 Carbon Neutral Strategy (Government of Korea, 2020)
	STI strategy	206.	Government R&D in the \$20 trillion eraGovernment R&DMid- to long-term investment strategy [정부R&D20조원시대이정부R&D중장기투자전략] (Ministry of Science and ICT, 2019)
		207.	2021 Action Plan for Science Gifted and Talented [2021년도 과학영재양성사업 시행계획] (Ministry of Science and ICT, 2021)
Luxembourg	Ministerial plan	208.	Impact: driving our excellence 2019, Annual Report [L'impact: le moteur de notre excellence, Rapport annuel 2019] (Luxembourg Institute of Science and Technology, 2020)
		209.	Driving change in a changing world, Annual Report 2020 [Vecteur de changement dans un monde en mutation, Rapport annuel 2020] (Luxembourg Institute of Science and Technology, 2021)
	Government strategy	210.	Luxembourg 2030 Third National Plan for Sustainable Development [Luxembourg 2030 3ème Plan National pour un Développement Durable] (Luxembourg 2030, 2019)
		211.	National Plan for Smart, Sustainable and Inclusive Growth of Luxembourg 2019 (Government of the Grand Duchy of Luxembourg, 2019)
		212.	National Plan for Smart, Sustainable and Inclusive Growth of Luxembourg 2020 (Government of the Grand Duchy of Luxembourg, 2020)
		213.	Plan for the Restart and Resilience of the Grand Duchy of Luxembourg [Plan pour la Reprise et la Résilience du Grand-Duché de Luxembourg] (Plan for Restart and Resilience of the Grand Duchy of Luxembourg, 2021)
		214.	National Plan for a Green,Digital and Inclusive Transition (Government of the Grand Duchy of Luxembourg, 2021)
	STI strategy	215.	The Data-Driven Innovation Strategy for the Development of a Trusted and Sustainable Economy in Luxembourg (Ministry of the Economy, 2019)
		216.	National Research and Innovation Strategy (Ministry of Higher Education and Research, 2020)
Netherlands	Ministerial plan	217.	Flywheel of Innovation in the Netherlands: Strategic Plan 2018-2021 (Netherlands Organisation for Applied Scientific Research, 2017)
		218.	Connecting Science and Society: NWO Strategy 2019-2022 (Netherlands Organisation for Applied Scientific Research, 2018)
		219.	Connecting, Changing, Accelerating: TNO Strategy 2022-2025 (Netherlands Organisation for Applied Scientific Research, 2021)
	Government strategy	220.	Parliament letter with Strategic Action Plan for Artificial Intelligence [Kamerbrief met Strategisch Actieplan voor Artificiële Intelligentie] (Ministry of Economic Affairs and Climate Policy, 2019)
		221.	Vision on Industry in the Netherlands (Government of the Netherlands, 2020)
		222.	National Reform Programme (Government of the Netherlands, 2020)
		223.	Integrated National Energy and Climate Plan (Ministry of Economic Affairs and Climate Policy, 2021)

	STI strategy	224.	Science Vision 2025, choices for the future [Wetenschapsvisie 2025, keuzes voor de toekomst] (Ministry of Education, Culture and Science, 2014)
		225.	Parliamentary letter on innovation policy and the promotion of innovation: towards mission-driven innovation policy with impact [Kamerbrief over innovatiebeleid en de bevordering van innovatie: naar missiegedreven innovatiebeleid met impact] (Ministry of Economic Affairs and Climate Policy, 2018)
		226.	Parliamentary letter on long-term growth strategy for the Netherlands [Kamerbrief over groeistrategie voor Nederland op de lange termijn] (Ministry of Economic Affairs and Climate Policy, 2019)
		227.	Dutch Strategy to Strengthen Research and Innovation Ecosystems (Ministry of Economic Affairs and Climate Policy, 2021)
New Zealand	Ministerial plan	228.	BIE Statement of Intent 2018–2022 (Ministry of Business, Innovation & Employment, 2018)
		229.	BIE 2018/19 Annual Report Growing New Zealand for All (Ministry of Business, Innovation & Employment, 2019)
		230.	Callaghan Innovation Annual Report 2019 (Callaghan Innovation, 2019)
		231.	BIE 2019/20 Annual Report Growing New Zealand for All (Ministry of Business, Innovation & Employment, 2020)
		232.	Callaghan Innovation Responding to a once-in-a-lifetime challenge (Callaghan Innovation, 2020)
	Government strategy	233.	National Statement of Science Investment 2015–2025 (Ministry of Business, Innovation & Employment, 2015)
		234.	A Refreshed Industry Strategy in Response to COVID-19 (Ministry of Business, Innovation & Employment, 2020)
	STI strategy	235.	New Zealand's Research, Science & Innovation Strategy: Draft for Consultation (New Zealand Government, 2019)
Slovakia	Government strategy	236.	Low-Carbon Development Strategy of the Slovak Republic until 2030 with a View to 2050 (Ministry of the Environment, 2017)
		237.	Strategy for Adaptation of the Slovak Republic to Climate Change [Stratégia Adaptácie Slovenskej Republiky na Zmenu Klímy] (Ministry of the Environment, 2018)
		238.	Action plan for the digital transformation of Slovakia for 2019 –2022 (Office of the Deputy Prime Minister for Investments and Information, 2018)
		239.	Strategy of the Digital Transformation of Slovakia 2030 (Office of the Deputy Prime Minister for Investments and Information, 2018)
		240.	Greener Slovakia: Strategy of the Environmental Policy of the Slovak Republic until 2030 (Ministry of the Environment, 2019)
		241.	National Reform Programme of the Slovak Republic 2019 [Národný program reforiem Slovenskej republiky 2019] (Government of Slovak Republic, 2019)
		242.	Integrated National Energy and Climate Plan 2021-2030 [Integrovaný národný energetický a klimatický plán roky 2021 - 2030] (Ministry for the Economy, 2019)
		243.	National Reform Programme of the Slovak Republic 2020 [Národný program reforiem Slovenskej republiky 2020] (Government of Slovak Republic, 2020)
		244.	Renewal Plan [Plán Obnovy] (Government of the Slovak Republic, 2021)
		245.	Stability programme of the Slovak Republic for the period 2021 to 2024 [Program stability Slovenskej republiky na roky 2021 až 2024] (Ministry of Finance, 2021)
	STI strategy	246.	Through knowledge towards prosperity -Research and Innovation Strategy for Smart Specialisation of the Slovak Republic (Ministry of Investment, Regional Development and Informatization of the Slovak Republic, 2013)
Spain	Ministerial plan	247.	Strategy Plan of CSIC 2018-2021 [Plan Estratégico del CSIC 2018-2021] (CSIC, 2018)
		248.	Annual Action Plan of the State Research Agency for the year 2020 [Plan de Acción Anual de la Agencia Estatal de Investigación para el año 2020] (Agencia Estatal de Investigación, 2020)
		249.	Annual Action Plan of the State Research Agency for the year 2021 [Plan de Acción Anual de la Agencia Estatal de Investigación para el año 2021] (Agencia Estatal de Investigación, 2021)
		250.	CSIC Action Plan 2021 [Plan de Acción CSIC 2021] (CSIC, 2021)
	Government strategy	251.	The Digital Transformation of the Spanish Industry [La Transformación Digital de la Industria Española] (Ministry of Economic Affairs and Digital Transformation, 2015)
		252.	National reform programme [Programa Nacional de Reformas] (Government of Spain, 2019)

		253.	Shock Plan for Science and Innovation [Plan de choque para la ciencia y la innovación] (Government of Spain, 2020)
		254.	National Artificial Intelligence Strategy [Estrategia Nacional de Inteligencia Artificial] (Ministerio de Asuntos Económicos y Transformación Digital, 2020)
		255.	Hydrogen Roadmap: a commitment to renewable hydrogen [Hoja de Ruta del Hidrógeno: una apuesta por el hidrógeno renovable] (2020)
		256.	National Integrated Energy Plan for Climate 2021 - 2030 [Plan Nacional Integrado de Energía y Clima 2021-2030] (Ministerio de Transición Ecológica, 2020)
		257.	National Reform Programme [Programa Nacional de Reformas] (Ministerio de Hacienda y Función Pública, 2020)
		258.	Recovery, Transformation, and Resilience Plan [Plan de Recuperación, Transformación y Resiliencia] (Government of Spain, 2021)
		259.	National Reform Programme [Programa Nacional de Reformas] (Ministerio de Hacienda y Función Pública, 2021)
	STI strategy	260.	State Plan for Scientific and Technical Research and Innovation [Plan Estatal de Investigación Científica y Técnica y de Innovación] (Ministerio de Ciencia e Innovación, 2018)
Sweden	Legislation	261.	Spanish Strategy for Science, Technology, and Innovation 2021-2027 [EECTI Estrategia Española de Ciencia, Tecnología e Innovación 2021-2027] (EECTI, 2020)
		262.	PEICTI State Plan for Scientific, Technical and Innovation Research 2021-2023 [PEICTI Plan Estatal de Investigación Científica, Técnica y de Innovación 2021-2023] (PEICTI, 2021)
	Government strategy	263.	Government Research Bill (Government of Sweden, 2016)
		264.	Government Research & Innovation Bill (Government of Sweden, 2020)
		265.	Roadmap for Fossil Free Competitiveness (Government of Sweden, 2019)
		266.	Sweden's National Reform Programme 2019 (Government of Sweden, 2019)
		267.	Sweden's Trade and Investment Strategy (Government of Sweden, 2019)
		268.	Circular Economy – Strategy for the Transition in Sweden (Government of Sweden, 2020)
		269.	Strategy for Fossil Fuel Competitiveness: Sustainable Value Battery Chain (Ministry of the Environment, 2020)
		270.	Sweden's National Reform Programme 2020 (Government of Sweden, 2020)
		271.	Sweden's recovery plan [Sveriges återhämtningsplan] (Government of Sweden, 2021)
		272.	A sustainable digitalized Sweden 2030 – a proposal for a strategic programme (Vinnova Swedish Innovation, 2021)
Switzerland	Ministerial plan	273.	Innosuisse Multiannual programme 2021-2024 [Innosuisse Programme pluriannuel 2021–2024] (Innosuisse, 2019)
		274.	Innosuisse Annual Report 2019 [Innosuisse Rapport de gestion 2019] (Innosuisse, 2020)
		275.	SNF Profile 2019-2020 [SNF Profile 2019-2020] (Swiss National Science Foundation, 2020)
		276.	Innosuisse Annual Report 2020 [Innosuisse Rapport de gestion 2020] (Swiss National Science Foundation, 2021)
	Government strategy	277.	Sustainable Development Strategy 2016-2019 [Stratégie pour le Développement Durable 2016–2019] (Federal Council, 2016)
		278.	Digital Switzerland Strategy [Stratégie Suisse numérique] (Federal Council, 2016)
		279.	2030 Sustainable Development Strategy [Stratégie pour le développement durable 2030] (Federal Council, 2021)
United Kingdom	STI strategy	280.	Switzerland's International Strategy for Education, Research and Innovation [Stratégie internationale de la Suisse dans le domaine formation, recherche et innovation] (Federal Council, 2018)
	Ministerial plan	281.	UKRI Annual Reports and Accounts 2018-19 (UK Research and Innovation, 2019)
		282.	UKRI Annual Reports and Accounts 2019-20 (UK Research and Innovation, 2020)
	Government strategy	283.	International Research and Innovation Strategy (HM Government, 2019)
		284.	Our plan to re-build: The UK Government's COVID-19 recovery strategy (HM Government, 2020)
		285.	Design in innovation strategy 2020-2024 (HM Government, 2020)
		286.	Covid-19 response? Spring 2021 (HM Government, 2021)

United States	STI strategy	287.	UK Plant Science Research Strategy: A Green Roadmap for the Next Ten Years (Professor Jane Langdale CBE FRS, University of Oxford, 2021)
		288.	Transforming our world with AI UKRI's role in embracing the opportunity (UK Research and Innovation, 2021)
		289.	UK Research and Development Roadmap (HM Government, 2020)
	Legislation	290.	Department of Energy Research and Innovation Act (115th Congress, 2018)
		291.	Coronavirus Aid, Relief and Economic Security Act (CARES) (116th Congress, 2020)
		292.	American Rescue Plan Act of 2021 (117th Congress, 2021)
	Ministerial plan	293.	Creating Technology Breakthroughs and New Capabilities for National Security, 2017 (Defense Advanced Research Projects Agency (DARPA), 2017)
		294.	Building the Future, Investing in Discovery and Innovation: NSF Strategic Plan for Fiscal Years 2018-2022 (National Science Foundation, 2018)
		295.	Creating Technology Breakthroughs and New Capabilities for National Security, 2019 (Defense Advanced Research Projects Agency (DARPA), 2019)
		296.	Advanced Research Projects Agency – Energy Annual Report for FY 2018 (Department of Energy, 2019)
		297.	NIH-wide Strategic Plan: Fiscal Years 2021-2025 (National Institutes of Health, 2020)
		298.	NIH-Wide Strategic Plan for COVID-19 Research (Updated) (National Institutes of Health, 2021)
	Government strategy	299.	Assessing and Strengthening the Manufacturing and Defense Industrial Base and Supply Chain Resiliency of the United States (Interagency Task Force in Fulfilment of Executive Order 13806, 2018)
		300.	National Strategic Computing Initiative Update: Pioneering the Future of Computing (National Science and Technology Council, 2019)
		301.	The National Artificial Intelligence Research and Development Strategic Plan: 2019 Update (National Science and Technology Council, 2019)
		302.	Pioneering the Future Advanced Computing Ecosystem: A Strategic Plan (National Science and Technology Council, 2020)
		303.	Hydrogen Program Plan (Department of Energy, 2020)
		304.	A Strategic Vision for America's Quantum Networks (The White House National Quantum Coordination Office, 2020)
		305.	Manufacturing USA Report to Congress Fiscal Year 2019 (Manufacturing USA, 2020)
		306.	Recommended Practices for Strengthening the Security and Integrity of America's Science and Technology Research Enterprise (National Science and Technology Council, 2021)
		307.	Final Report of the National Security Commission on Artificial Intelligence (National Security Commission on Artificial Intelligence, 2021)
		308.	Building Resilient Supply Chains, Revitalizing American Manufacturing, and Fostering Broad-Based Growth (White House, 2021)
		309.	Manufacturing USA Report to Congress Fiscal Year 2020 Including COVID-19 Emergency Response Highlights (Manufacturing USA, 2021)
	STI strategy	310.	Fiscal Year 2021 Administration Research and Development Budget Priorities (Office of Management and Budget and Office of Science and Technology Policy, 2019)
		311.	Fiscal Year (FY) 2022 Administration Research and Development Budget Priorities and Cross-cutting Actions (Office of Management and Budget and Office of Science and Technology Policy, 2020)
		312.	Multi-Agency Research and Development Priorities for the FY 2023 Budget (Executive Office of the President, 2021)
		313.	A Letter to Dr. Eric S. Lander, the President's Science Advisor and Nominee as Director of the Office of Science and Technology Policy (White House, 2021)